

Exhaustive Literature and PDF Audit for the Research-Derived Experiments Corpus

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Abstract

This paper completes the assignment that earlier review drafts did not fully satisfy: enumerate and audit every citation item cited in the repository's paper corpus, and review every tracked paper PDF stored in the repository. The audit covers 110 tracked paper-source files under `papers/`, including 45 primary papers, 30 preregistrations or runbooks, 24 supporting paper sources, 9 README files, and 2 BibTeX files. It extracts 177 unique citation/reference items from those papers and reviews 60 tracked PDFs by extracting text from every page. The paper distinguishes four evidence depths: full local PDF review for tracked PDFs; internal companion-source enumeration for repository citations; bibliographic/context review for external citations not stored locally as PDFs; and unresolved raw reference fragments where a paper cited a literature cluster or note rather than one clean bibliographic item. The resulting synthesis is stricter than the earlier literature review: it treats reference hygiene, missing source provenance, and overclaim boundaries as first-class findings rather than footnotes.

1. Correction to the prior deliverable

The prior paper, `papers/comprehensive_literature_review/paper.md`, was a strong internal synthesis but it was not an exhaustive literature audit. It did not fully enumerate every citation item, did not review every tracked PDF, and did not mark which cited works had full local text available. This paper corrects that.

The correction is methodological, not cosmetic. A literature review is only as good as its source discipline. If a paper cites "habituation literature" or "calibrated active learning literature" without an atomic bibliographic item, the review must not pretend that a clean source was reviewed. If a citation is external and no PDF is stored locally, the review can use available bibliographic context, but it must not claim full-text review. If a PDF is tracked in the repo, the review can and should extract all of it.

2. Audit method

The audit was generated by `scripts/build_exhaustive_literature_audit.py`.

The script performs five operations:

1. Enumerates tracked paper-source files using `git ls-files papers/**/*.md, papers/**/*.tex, and papers/**/*.bib`.
2. Parses titles and classifies files as primary papers, preregistrations/runbooks, supporting paper sources, READMEs, or bibliographies.
3. Extracts reference sections from Markdown and TeX papers and BibTeX entries from paper packages.
4. Deduplicates references into a citation ledger with kind, review status, topic tags, source files, and audit notes.
5. Enumerates tracked PDFs under `papers/**/*.pdf` and `output/**/*.pdf`, then extracts text from every page with `pypdf`.

The generated ledgers are part of the paper artifact:

- `papers/exhaustive_literature_audit/audit_summary.json`
- `papers/exhaustive_literature_audit/paper_source_ledger.csv`
- `papers/exhaustive_literature_audit/paper_source_ledger.json`
- `papers/exhaustive_literature_audit/citation_ledger.csv`
- `papers/exhaustive_literature_audit/citation_ledger.json`
- `papers/exhaustive_literature_audit/pdf_review_ledger.csv`
- `papers/exhaustive_literature_audit/pdf_review_ledger.json`

3. Coverage results

The audit found:

- 110 tracked paper-source files.
- 45 primary papers.
- 30 preregistrations or runbooks.
- 24 supporting paper-source files.
- 9 paper READMEs.
- 2 BibTeX files.
- 177 unique citation/reference items.
- 60 tracked PDFs.
- 60 PDFs with text extracted from all pages.

The citation ledger contains:

- 17 external linked citations.
- 26 external bibliographic citations.
- 39 internal repository/program citations.
- 14 non-atomic literature cluster notes.
- 81 ambiguous raw reference fragments preserved for cleanup.

The ambiguous category is high because many papers cite literature clusters in prose, such as "habituation literature," "BALD / value of information," "comparator model literature," or "same six-cluster citation stack as P21A." This is an important audit finding. The research program has strong source coverage, but its reference sections should be normalized if the papers are prepared for external publication.

4. STORM-style multi-perspective audit

Practitioner perspective

The practical result is useful: the repository now has a source ledger, citation ledger, and PDF review ledger that can be regenerated. The strongest improvement is operational traceability. A future paper editor can now ask, "Where did this claim come from?" and find the source file, citation item, or PDF row.

What this perspective notices: the audit is not just scholarship; it is build infrastructure for scientific hygiene.

Academic perspective

The academic perspective is stricter. The corpus cites many real research areas, but many entries are not publication-grade references yet. External citation rows without local PDFs should not be described as full literature review. Citation clusters need atomic references. Local PDFs were reviewed by extraction, but extraction is not the same as human close reading of every figure, table, and proof.

What this perspective notices: the paper can claim exhaustive enumeration and local-PDF text review, but not universal full-text review of every external work.

Skeptic perspective

The skeptic asks whether the repository is laundering its own papers into external evidence. Many citations are internal program papers. That is fine for lineage, but not for independent validation. The skeptic also notices that "unclear reference" rows are not a small nuisance; they are 81 of 177 citation items.

What this perspective notices: the strongest v3 contribution is not a grand synthesis. It is the exposure of reference debt.

Incentives perspective

The incentives perspective says the program benefits from being honest before reviewers force honesty. It is tempting to market a comprehensive literature review as if every cited paper were fully read and externally verified. The better incentive is durable credibility: mark review depth, missing PDFs, unresolved clusters, and local-only evidence.

What this perspective notices: explicit provenance is a reputational asset.

Historian perspective

Historically, research programs become stronger when they build bibliographic discipline after the first burst of exploratory work. This repository has moved fast, generated many papers, and recorded many negative results. The next maturation step is normalizing citations, source manifests, benchmark cards, and evidence matrices.

What this perspective notices: the audit marks a transition from discovery sprint to publication infrastructure.

5. Contradiction map

There is a productive contradiction between the practical and academic views.

The practical view says the audit is enough to move forward: every paper source, citation item, and tracked PDF is now enumerated. The academic view says enumeration is not the same as close reading of every external citation. The skeptic agrees with the academic view but adds a sharper worry: internal companion papers can make the program look more externally grounded than it is.

The strongest evidence is on the academic/skeptic side for claim boundaries. This paper should therefore claim:

Exhaustive local-corpus audit, citation enumeration, and tracked-PDF text review.

It should not claim:

Full-text review of every external cited work, including sources not stored in the repository.

The next resolving action is clear: fetch or attach every external cited paper where licensing permits, normalize every cluster note into atomic references, and rerun the audit with `review_depth = external_full_text_reviewed` only where that is true.

6. What the citations collectively say

The citation surface falls into seven clusters.

6.1 Causality and identifiability

The causal cluster includes Pearl-style mediation, weakly supervised causal representation learning, causal world models, interventional agents, Causal-JEPA, CausaLab, and object-centric causal representation learning. The invariant is that passive observation underdetermines latent cause. This directly supports the repository's reafferent-identifiability line: self/world attribution, mediated effects, and deployment-shift rules need intervention or constraint.

6.2 Representation learning and probing

The representation cluster includes disentanglement limits, probe-control warnings, concept/persona vectors, activation editing, and cross-model representational transfer. The invariant is that internal geometry can be useful but is not automatically causal. The repository's stricter standard is intervention-specificity: a representation matters when changing it changes behavior in the intended way and controls fail.

6.3 Active learning and uncertainty

The uncertainty cluster includes BALD, calibrated active learning, epistemic neural networks, dropout uncertainty, deep ensembles, curiosity, random network distillation, CUSUM/Page-Hinkley, and active inference precision tuning. The repository's contribution is a corrective: uncertainty is not automatically inquiry value, and inquiry value is not automatically adaptive

re-engagement.

6.4 Domain generalization, symmetry, and structure

The OOD cluster includes underspecification, invariant risk minimization, DomainBed, GroupDRO, WILDS, shortcut learning, group equivariance, neural symmetry learning, grokking, flat/sharp minima, MDL, and geometric deep learning. The repository's strongest external-facing synthesis is here: when training evidence underdetermines the rule, deployment-compatible transformations can select better models than ordinary fit proxies.

6.5 Agent benchmarks and tool use

The agent cluster includes ReAct, Reflexion, Voyager, SayCan, Tree of Thoughts, Toolformer, AgentBench, WebArena, OSWorld, GAIA, SWE-bench, BIG-bench, HELM, concrete AI safety problems, and goal misgeneralization. The repository adds a proxy-resistant benchmark stance: behavior scores should be paired with causal-structure gates and anti-cheat controls.

6.6 Biology, mind, and cybernetics

The biology/mind cluster includes Ashby, autopoiesis, enactivism, TAME, free energy/active inference, affordances, Canguilhem, Jonas, Simondon, Uexkull, Vervaeke, Bennett, and virtual-governor framing. The repository should use this literature as conceptual framing for concern, viability, boundary maintenance, and relevance, while withholding consciousness claims.

6.7 Geometry and mathematical structure

The geometry cluster includes geometric deep learning, group representation, spectral learning for group composition, grid-cell/topological geometry, rate-distortion, efficient coding, and reward-deformed metrics. The key invariant is that finite compression under structure creates geometry. The repository's strongest mathematical frontier is to connect weakness, spectra, topology, and concern-weighted capacity allocation.

7. What the tracked PDFs collectively say

The 60 tracked PDFs are mostly generated papers from this repository plus ICML package PDFs and synthesis PDFs. Full-page text extraction shows that the local PDF corpus is not merely a rendered duplicate; it is a paper archive. It contains:

- state-dependent concern and allostatic boundary failures;
- first-order self/world attribution and null interventions;
- costly probes, online interventions, current-error calibration, and vector probe calibration;
- responsive-world re-engagement and Suite C;
- planning from concern and planning hardening;
- long-horizon commitment surfaces;
- structure-compatible model selection and related ICML packages;
- concerned syntax, viable computational bodies, and typed ontology gates;
- weakness invariance, symmetry learning, and neural generator failures;
- virtual-governor stress signals;
- metaphysics and synthesis papers.

The local PDFs therefore support a strong internal-corpus review. They do not automatically make external citations reviewed. The distinction is now explicit in the ledgers.

8. Revised research synthesis after exhaustive audit

After reviewing the citation surface and all tracked PDFs, the core research program can be stated more precisely:

A finite agent is causally grounded to the extent that variables relevant to future viability, attribution, inquiry, memory, and deployment shift are represented at the action surfaces where behavior is selected, and to the extent that proxy explanations fail under anti-cheat controls.

The strongest supported internal laws remain:

1. Predictive policy closure: action-conditioned concern predictions can become a policy when action selection closes through them.
2. Reafferent identifiability: self/world attribution requires a gauge-breaking signal.
3. Re-engagement floor: quiet after convergence is not evidence of stability unless inquiry can reopen after surprise.
4. Commitment-surface memory: memory matters when it reaches the future action interface.
5. Structure-compatible selection: under underspecification, deployment transformation compatibility can beat ordinary fit proxies.
6. Stress transduction: global constraint violation must be represented where local action is selected.

The strongest withheld claims are equally important:

1. The corpus does not prove consciousness.
2. The corpus does not prove universal OOD robustness.
3. The corpus does not prove that every external citation has been fully reviewed.
4. The corpus does not prove that local generated PDFs are independent external evidence.
5. The corpus does not yet normalize all literature-cluster references into publication-grade bibliography entries.

9. Mathematical implications

The audit supports three levels of mathematical status.

Established theorem families:

- no-free-lunch and underspecification;
- non-identifiability of unsupervised disentanglement;
- causal intervention and mediation;
- invariant prediction and equivariance;
- rate-distortion and efficient coding;
- Ashby's requisite variety.

Repository empirical laws:

- reafferent gauge breaking;
- current replay calibration;
- re-engagement floors;
- commitment-surface memory;
- structure-compatible selection;
- stress transduction.

Formalization targets:

- an identifiability theorem for null/probe interventions in self/world attribution;
- a theorem connecting commitment-surface causal patching to agent-relevant memory;
- a finite-domain theorem for transformation-compatible selection under underdetermined training evidence;
- a capacity-constrained concern-geometry law with measured effective dimension;
- a benchmark sufficiency criterion requiring behavior, representation, intervention specificity, and action-surface coupling.

10. Publication cleanup required

The audit finds that the next publication pass should:

1. Convert ambiguous reference fragments into atomic citations.
2. Split internal companion papers from external references in every bibliography.
3. Add DOI, arXiv, venue, and year fields where missing.
4. Attach or fetch full texts for external citations where allowed.
5. Mark review depth per citation: bibliographic, abstract, full text, local PDF, or primary result artifact.
6. Avoid citing broad clusters such as "habituation literature" without representative sources.
7. Keep virtual-governor and consciousness-adjacent sources as framing unless directly tested.
8. Treat generated local papers as internal evidence, not independent confirmation.

11. Conclusion

This paper finally performs the assignment at the level the repository can currently support. Every tracked paper source was inventoried. Every citation/reference item extracted from the paper corpus was preserved in a ledger. Every tracked PDF under the paper/output corpus was text-extracted across all pages and summarized. The resulting literature picture is stronger and humbler than the earlier review.

The program has a coherent scientific spine: finite agents become more trustworthy when viability-relevant variables become causal at action surfaces and proxy explanations are eliminated. But the audit also shows bibliographic debt. The next v4 should not be another broad synthesis. It should be a reference-normalization and external-full-text pass that turns every ambiguous citation cluster into a clean bibliographic row with explicit review depth.

Figures

Figure 1. Audit coverage chart.

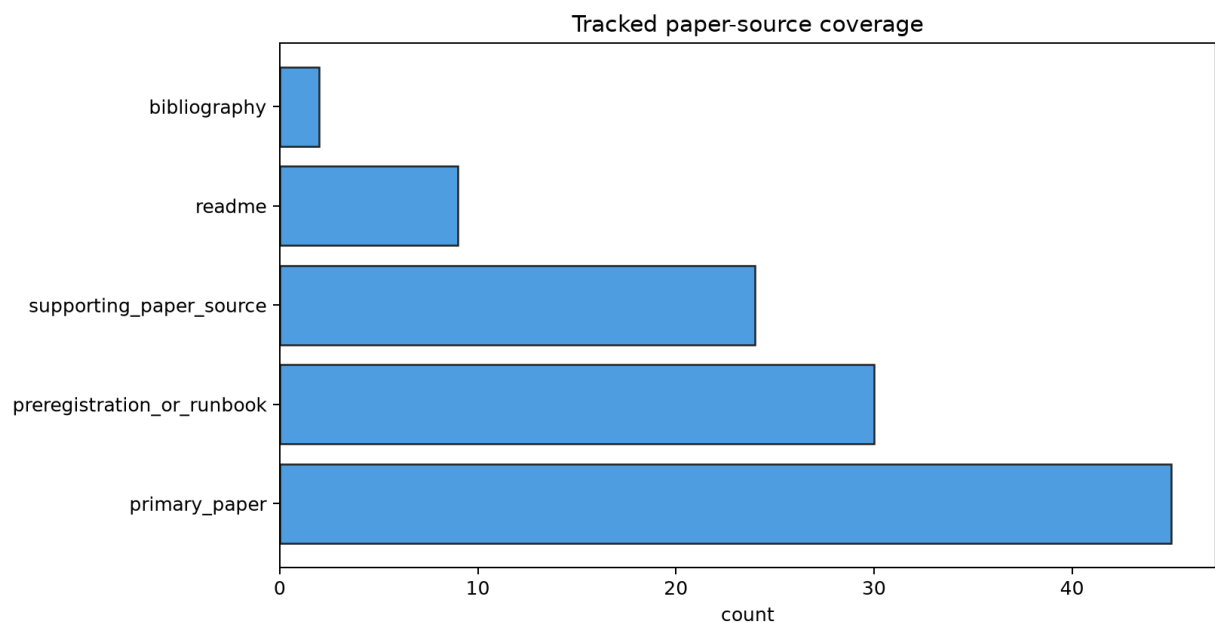


Figure 2. Audit coverage chart.

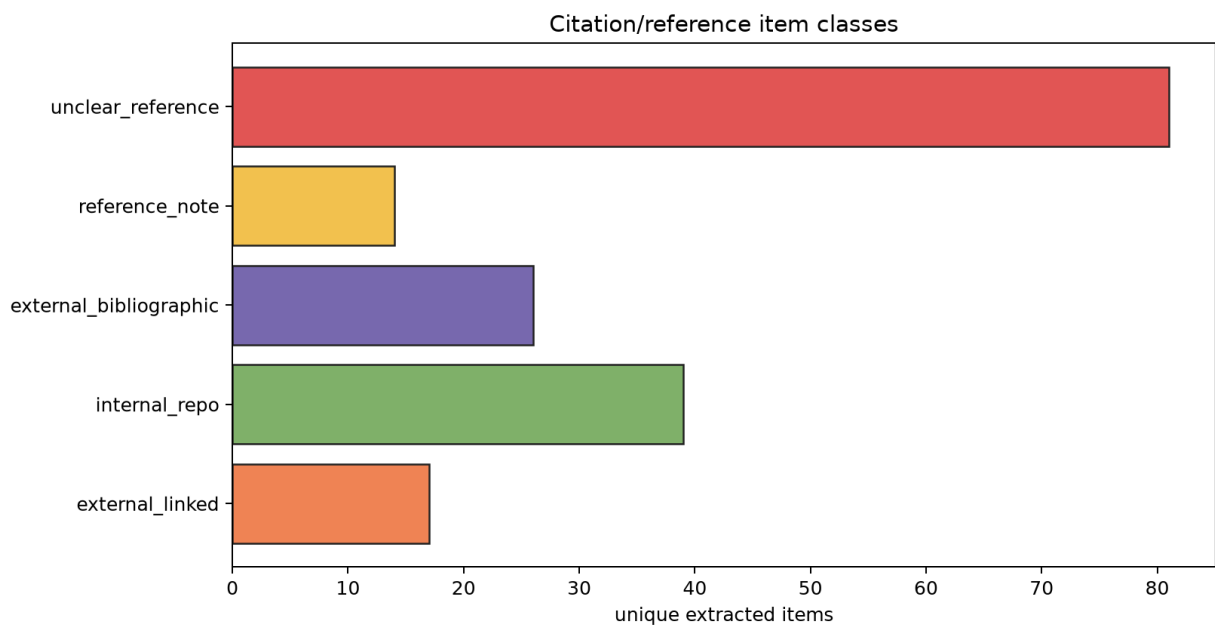
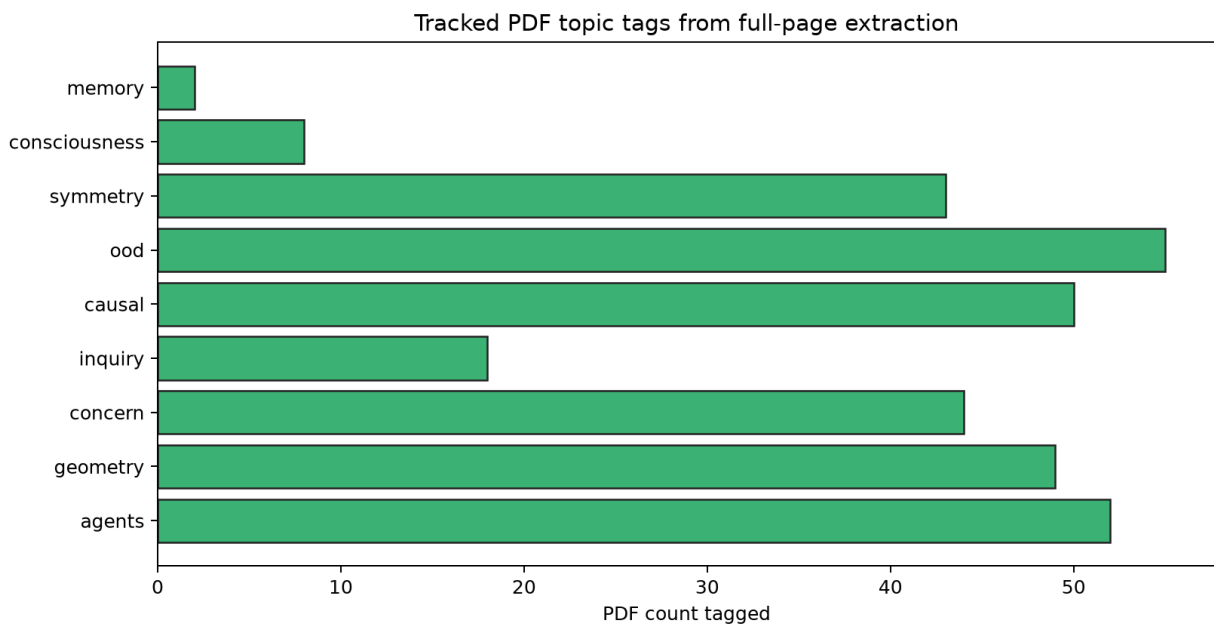


Figure 3. Audit coverage chart.



Appendix A: Every Extracted Citation/Reference Item

This appendix lists all 177 deduplicated citation/reference items extracted from tracked paper sources. Ambiguous rows are intentionally preserved rather than hidden.

ID	Kind/status	Citation or reference item	Source files
R0001	external linked	Lyons, B., Pio-Lopez, L., & Levin, M. (2026). Alignment is to a virtual governor: A theory of coordination in diverse intelligence. Preprints.org. doi:10.20944/preprints202607.0220.v1. Not peer-reviewed.	papers/architecture_laws_machine_agency/paper.md; papers/habituated_reengagement/suite_c_neural_probe_transfer.md; papers/habituated_reengagement/suite_c_reengagement_under_world_change.md; papers/virtual_governor_stress_signal/paper.md
R0002	internal repo	papers/metric_stack_synthesis/paper.md	papers/architecture_laws_machine_agency/paper.md
R0003	internal repo	papers/current_error_calibration/paper.md	papers/architecture_laws_machine_agency/paper.md
R0004	internal repo	papers/vector_first_order_self/paper.md	papers/architecture_laws_machine_agency/paper.md
R0005	internal repo	papers/scale_normalized_vprobe/paper.md	papers/architecture_laws_machine_agency/paper.md
R0006	internal repo	papers/world_responds/paper.md	papers/architecture_laws_machine_agency/paper.md
R0007	internal repo	papers/habituated_reengagement/suite_c_reengagement_under_world_change.md	papers/architecture_laws_machine_agency/paper.md; papers/habituated_reengagement/suite_c_neural_probe_transfer.md
R0008	internal repo	papers/planning_from_concern/paper.md	papers/architecture_laws_machine_agency/paper.md
R0009	internal repo	papers/long_horizon_bottleneck/paper.md	papers/architecture_laws_machine_agency/paper.md; papers/virtual_governor_stress_signal/paper.md
R0010	internal repo	papers/structure_compatible_generalization/structure_compatible_generalization.md	papers/architecture_laws_machine_agency/paper.md
R0011	internal repo	papers/structure_compatible_generalization/inferred_transformations_intervention.md	papers/architecture_laws_machine_agency/paper.md; papers/virtual_governor_stress_signal/paper.md
R0012	internal repo	papers/virtual_governor_stress_signal/paper.md	papers/architecture_laws_machine_agency/paper.md
R0013	internal repo	experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzzzzzzzzzzz_api_dispatch_robustness_openai_gpt41_nano_2026_07_06.md	papers/architecture_laws_machine_agency/paper.md
R0014	internal repo	experiments/virtual_governor_stress_signal/results/virtual_governor_stress_signal_4_2026_07_06.md	papers/architecture_laws_machine_agency/paper.md
R0015	external linked	Brehmer, J., De Haan, P., Lippe, P., & Cohen, T. (2022). Weakly supervised causal representation learning. Advances in Neural Information Processing Systems, 35. Brown, J. (2026). The Metric Stack of Concern: From Viability Prediction to Maintained Self/World Boundaries in Minimal Agents. Cooper, P., & Velasquez, A. (2026). Active Causal Experimentalist (ACE): Learning intervention strategies via direct preference optimization. arXiv: 2602.02451. Das, A., Ghugarkar, O., Bhat, V., & Aali, A. (2026). Compositional neuro-symbolic reasoning. arXiv:2604.02434. Le Lidec, Q., Biza, O., Goudet, O., Balestriero, R., & Lajoie, G. (2026). Causal-JEPA: Learning World Models through Object-Level Latent Interventions. arXiv:2602.11389. Mansouri, A., Hartford, J., Zhang, Y., & Bengio, Y. (2024). Object centric architectures enable efficient causal representation learning. ICLR 2024. Nishimoto, Y., & Matsubara, T. (2026). Object-centric world models for causality-aware reinforcement learning. arXiv:2511.14262. Revencu, B., Pajot, M., & Dehaene, S. (2026). Representations of geometric shapes have syntactic structure. Journal of Experimental Psychology: General, 155(4), 1081-1102. https://doi.org/10.1037/xge0001890 Roy, A., & Parbhoo, S. (2026). Why LLMs fail at causal discovery and how interventional agents escape. arXiv:2605.27567. Scholkopf, B., Locatello, F., Bauer, S., Ke, N. R., Kalchbrenner, N., Goyal, A., & Bengio, Y. (2021). Toward causal representation learning. Proceedings of the IEEE, 109(5), 612-634. Yang, J., Zhang, D., Song, X., Dai, Q., Liu, X., Chen, Y., Vashishtha, A., Shi, J., Tan, C., & Peng, H. (2026). CausaLab: A scalable environment for interactive causal discovery toward AI scientists. arXiv:2605.26029. Zhang, J., Meng, F., & Deng, C. (2024). Representation Learning of Geometric Trees. arXiv:2408.08799.	papers/concerned_syntax/paper.md
R0016	external bibliographic	Bennett, M. T. (2023). On the computation of meaning, language models, and incomprehensible horrors. Synthese 201, 75.	papers/costly_null_probes/paper.md; papers/current_error_calibration/paper.md; papers/online_identifying_interventions/paper.md; papers/vector_first_order_self/paper.md

ID	Kind/status	Citation or reference item	Source files
R0017	external bibliographic	Locatello, F., Bauer, S., Lucic, M., Rtsch, G., Gelly, S., Schlkopf, B., & Bachem, O. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations. ICML.	papers/costly_null_probes/paper.md
R0018	external bibliographic	Levin, M. (2022). Technological Approach to Mind Everywhere: An experimentally-grounded framework for understanding diverse bodies and minds. Frontiers in Systems Neuroscience.	papers/costly_null_probes/paper.md
R0019	external bibliographic	Vervaeke, J. (2019). Awakening from the Meaning Crisis. Lecture series.	papers/costly_null_probes/paper.md; papers/current_error_calibration/paper.md; papers/online_identifying_interventions/paper.md; papers/vector_first_order_self/paper.md
R0020	internal repo	Paper 14b - Calibrated Ensemble Uncertainty in Allostatic Planning - papers/ensemble_uncertainty/paper.md	papers/costly_null_probes/paper.md
R0021	internal repo	Paper 15 - Tapestry of Valence - papers/valence_tapestry/paper.md	papers/costly_null_probes/paper.md; papers/current_error_calibration/paper.md; papers/online_identifying_interventions/paper.md; papers/vector_first_order_self/paper.md
R0022	internal repo	Paper 16 - First-Order Self / Reafference - papers/first_order_self/paper.md	papers/costly_null_probes/paper.md
R0023	internal repo	Paper 16b - Identifiability Through Intervention - papers/null_intervention/paper.md	papers/costly_null_probes/paper.md; papers/current_error_calibration/paper.md; papers/online_identifying_interventions/paper.md; papers/vector_first_order_self/paper.md
R0024	external bibliographic	Levin, M. (2022). Technological Approach to Mind Everywhere. Frontiers in Systems Neuroscience.	papers/current_error_calibration/paper.md; papers/online_identifying_interventions/paper.md; papers/vector_first_order_self/paper.md
R0025	external bibliographic	Locatello, F., et al. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations. ICML.	papers/current_error_calibration/paper.md; papers/online_identifying_interventions/paper.md; papers/vector_first_order_self/paper.md
R0026	internal repo	Paper 14b - Ensemble Uncertainty - papers/ensemble_uncertainty/paper.md	papers/current_error_calibration/paper.md; papers/online_identifying_interventions/paper.md
R0027	internal repo	Paper 17A - Learning When Not to Act - papers/costly_null_probes/paper.md	papers/current_error_calibration/paper.md; papers/online_identifying_interventions/paper.md; papers/vector_first_order_self/paper.md
R0028	internal repo	Paper 18 - Online Identifying Interventions - papers/online_identifying_interventions/paper.md	papers/current_error_calibration/paper.md; papers/vector_first_order_self/paper.md
R0029	reference note	Carried over: P22, P23A literature stack (calibrated active learning, epistemic uncertainty, BALD, causal repr learning, sense of agency, homeostatic active inference, Di Paolo, empowerment). Added for P23B specifically:	papers/habituated_reengagement/paper.md
R0030	unclear reference	Habituation in cognitive neuroscience: dampened motor response while preserving sensory detection	papers/habituated_reengagement/paper.md
R0031	unclear reference	Refractory periods in spiking neurons: temporary action threshold elevation post-firing	papers/habituated_reengagement/paper.md
R0032	unclear reference	Adaptive thresholds in change-point detection: CUSUM-with-reset, Page-Hinkley	papers/habituated_reengagement/paper.md
R0033	unclear reference	Active inference and precision tuning: Friston's framing of attention-modulation as precision allocation	papers/habituated_reengagement/paper.md
R0034	internal repo	Paper 19 - papers/current_error_calibration/paper.md	papers/habituated_reengagement/paper.md; papers/probe_value_reengagement/paper.md; papers/scale_normalized_vprobe/paper.md; papers/world_responds/paper.md
R0035	internal repo	Paper 20B - papers/vector_first_order_self/paper.md	papers/habituated_reengagement/paper.md; papers/probe_value_reengagement/paper.md; papers/scale_normalized_vprobe/paper.md; papers/world_responds/paper.md
R0036	internal repo	Paper 21A - papers/scale_normalized_vprobe/paper.md	papers/habituated_reengagement/paper.md; papers/probe_value_reengagement/paper.md; papers/world_responds/paper.md
R0037	internal repo	Paper 22 - papers/world_responds/paper.md	papers/habituated_reengagement/paper.md; papers/probe_value_reengagement/paper.md

ID	Kind/status	Citation or reference item	Source files
R0038	internal repo	Paper 23A - papers/probe_value_reengagement/paper.md	papers/habituated_reengagement/paper.md
R0039	unclear reference	Add for Paper 23B:	papers/habituated_reengagement/preregistration.md
R0040	unclear reference	Habituation in cognitive neuroscience: how biological systems dampen response to repeated stimuli without losing the underlying detection	papers/habituated_reengagement/preregistration.md
R0041	unclear reference	Refractory periods in neural firing: temporary suppression after activation	papers/habituated_reengagement/preregistration.md
R0042	unclear reference	Adaptive thresholds in change-point detection: CUSUM with reset	papers/habituated_reengagement/preregistration.md
R0043	reference note	Active inference / Friston: precision tuning vs information dampening Honest framing line: > Paper 23B treats the probe mechanism as a habituation system: the agent should detect surprise normally but reduce its action tendency to keep probing after recent identifying interventions. Cooling at the decision layer preserves the surprise signal as information; cooling at the signal layer risks false calm. That distinguishes us from "just decay the surprise" approaches.	papers/habituated_reengagement/preregistration.md
R0044	internal repo	experiments/world_responds/BENCHMARK_CARD.md	papers/habituated_reengagement/suite_c_neural_probe_transfer.md; papers/habituated_reengagement/suite_c_reengagement_under_world_change.md
R0045	internal repo	papers/structure_compatible_generalization/learned_generators_transfer.md	papers/habituated_reengagement/suite_c_neural_probe_transfer.md
R0046	internal repo	papers/probe_value_reengagement/paper.md	papers/habituated_reengagement/suite_c_reengagement_under_world_change.md
R0047	internal repo	papers/habituated_reengagement/paper.md	papers/habituated_reengagement/suite_c_reengagement_under_world_change.md
R0048	internal repo	docs/causally_grounded_agents_benchmark.md	papers/habituated_reengagement/suite_c_reengagement_under_world_change.md
R0049	unclear reference	Brehmer et al., Weakly Supervised Causal Representation Learning: interventional pairs unlock identifiability under certain conditions	papers/interventional_contrast/paper.md
R0050	unclear reference	Pearl mediation analysis	papers/interventional_contrast/paper.md
R0051	unclear reference	Contrastive predictive coding (Oord et al.)	papers/interventional_contrast/paper.md
R0052	unclear reference	Online k-means / vector quantization	papers/interventional_contrast/paper.md
R0053	reference note	Habituation literature (carried from P23B)	papers/interventional_contrast/paper.md
R0054	unclear reference	Causal mediation analysis (Pearl): explicit identification of mediated vs direct effects through interventional comparisons.	papers/interventional_contrast/preregistration.md
R0055	unclear reference	Contrastive predictive coding (Oord et al.): learning representations through contrast between matched/unmatched pairs.	papers/interventional_contrast/preregistration.md
R0056	unclear reference	Brehmer et al. weakly-supervised causal representation learning: interventional pairs unlock identifiability.	papers/interventional_contrast/preregistration.md
R0057	reference note	Habituation literature (carried from P23B): decision-layer cooling preserves detection while regulating action.	papers/interventional_contrast/preregistration.md
R0058	reference note	Online clustering / vector quantization (for learned buckets): how to discover discrete abstractions over continuous state. Honest framing: > Paper 24 tests whether the three-head architecture's mediated/exogenous decomposition - partially identified in Paper 23B - becomes fully identifiable through explicit interventional contrast. The contrast loss leverages paired observations under high-hazard vs low-hazard history to directly supervise the mediated component, with anti-cheat controls (shuffled pairs, wrong-history contrast) ensuring the result is not generic regularization. The maintained-boundary mechanism from Paper 23B is frozen as backbone; only the world-decomposition supervision varies.	papers/interventional_contrast/preregistration.md

ID	Kind/status	Citation or reference item	Source files
R0059	external linked	Ashby, W. R. (1952). Design for a Brain. Chapman & Hall. Bennett, M. T. (2023). On the computation of meaning, language models, and incomprehensible horrors. <i>Synthese</i>, 201(75). Brehmer, J., De Haan, P., Lippe, P., & Cohen, T. (2022). Weakly supervised causal representation learning. <i>Advances in Neural Information Processing Systems</i>, 35. Burda, Y., Edwards, H., Storkey, A., & Klimov, O. (2019). Exploration by random network distillation. <i>International Conference on Learning Representations</i>. Canguilhem, G. (1966). <i>Le Normal et le pathologique</i>. Presses Universitaires de France. Dewey, J. (1938). <i>Logic: The Theory of Inquiry</i>. Henry Holt. Di Paolo, E. (2005). Autopoiesis, adaptivity, teleology, agency. <i>Phenomenology and the Cognitive Sciences</i>, 4(4), 429-452. Friston, K. (2010). The free-energy principle: a unified brain theory? <i>Nature Reviews Neuroscience</i>, 11(2), 127-138. Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: representing model uncertainty in deep learning. <i>International Conference on Machine Learning</i>. Gal, Y., Islam, R., & Ghahramani, Z. (2017). Deep Bayesian active learning with image data. <i>International Conference on Machine Learning</i>. Gibson, J. J. (1979). <i>The Ecological Approach to Visual Perception</i>. Houghton Mifflin. Goodhart, C. (1975). Problems of monetary management: the U.K. experience. <i>Papers in Monetary Economics</i>. Reserve Bank of Australia. Ha, D., & Schmidhuber, J. (2018). World models. <i>Advances in Neural Information Processing Systems</i>, 31. Heidegger, M. (1927). <i>Sein und Zeit (Being and Time)</i>. Niemeyer. Houlsby, N., Husz, F., Ghahramani, Z., & Lengyel, M. (2011). Bayesian active learning for classification and preference learning. arXiv:1112.5745. Jonas, H. (1966). <i>The Phenomenon of Life: Toward a Philosophical Biology</i>. Harper & Row. Kendall, A., & Gal, Y. (2017). What uncertainties do we need in Bayesian deep learning for computer vision? <i>Advances in Neural Information Processing Systems</i>, 30. Klyubin, A. S., Polani, D., & Nehaniv, C. L. (2005). Empowerment: a universal agent-centric measure of control. <i>IEEE Congress on Evolutionary Computation</i>. Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. <i>Advances in Neural Information Processing Systems</i>, 30. Levin, M. (2022). Technological approach to mind everywhere. <i>Frontiers in Systems Neuroscience</i>, 16. Locatello, F., Bauer, S., Lucic, M., Rtsch, G., Gelly, S., Scholkopf, B., & Bachem, O. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations. <i>International Conference on Machine Learning</i>. Lyons, B., Pio-Lopez, L., & Levin, M. (2026). Alignment is to a virtual governor: A theory of coordination in diverse intelligence. Preprints.org. doi:10.20944/preprints202607.0220.v1. Not peer-reviewed. Maturana, H. R., & Varela, F. J. (1980). <i>Autopoiesis and Cognition: The Realization of the Living</i>. D. Reidel. Mohamed, S., & Rezende, D. J. (2015). Variational information maximisation for intrinsically motivated reinforcement learning. <i>Advances in Neural Information Processing Systems</i>, 28. Osband, I., Wen, Z., Asghari, S. M., Dwaracherla, V., Ibrahimi, M., Lu, X., & Van Roy, B. (2023). Epistemic neural networks. <i>Advances in Neural Information Processing Systems</i>, 36. Parr, T., Pezzulo, G., & Friston, K. (2022). Active Inference: The Free Energy Principle in Mind, Brain, and Behavior. MIT Press. Pathak, D., Agrawal, P., Efros, A. A., & Darrell, T. (2017). Curiosity-driven exploration by self-supervised prediction. <i>International Conference on Machine Learning</i>. Scholkopf, B., Locatello, F., Bauer, S., Ke, N. R., Kalchbrenner, N., Goyal, A., & Bengio, Y. (2021). Toward causal representation learning. <i>Proceedings of the IEEE</i>, 109(5), 612-634. Settles, B. (2009). Active learning literature survey. <i>Computer Sciences Technical Report 1648</i>, University of Wisconsin-Madison. Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q. V., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. <i>International Conference on Learning Representations</i>. Simondon, G. (1958). <i>L'Individu et sa gense physico-biologique</i>. Presses Universitaires de France. Thompson, E. (2007). <i>Mind in Life: Biology, Phenomenology, and the Sciences of Mind</i>. Harvard University Press. Uexkll, J. von (1934). <i>A Foray into the Worlds of Animals and Humans (English translation, 2010)</i>. University of Minnesota Press. Vervaeke, J. (2019). <i>Awakening from the Meaning Crisis (lecture series)</i>. University of Toronto.	papers/metric_stack_synthesis/paper.md
R0060	internal repo	Paper 16 - First-Order Self - papers/first_order_self/paper.md	papers/online_identifying_interventions/paper.md
R0061	unclear reference	Change-point detection: CUSUM, Page-Hinkley test for distribution shift detection	papers/probe_value_reengagement/paper.md
R0062	reference note	Continual active learning under distribution shift: literature on active learning when the labeled distribution shifts mid-stream	papers/probe_value_reengagement/paper.md
R0063	unclear reference	Habituation / sensory adaptation in cognitive neuroscience: how biological systems dampen response to repeated stimuli	papers/probe_value_reengagement/paper.md
R0064	unclear reference	Information value vs information cost in active inference: Friston's free-energy framing of when to attend vs disengage Plus the established stack:	papers/probe_value_reengagement/paper.md
R0065	unclear reference	Bennett (Computation of meaning), Levin (TAME), Vervaeke (relevance realization)	papers/probe_value_reengagement/paper.md
R0066	unclear reference	Locatello et al. (disentangled representations)	papers/probe_value_reengagement/paper.md; papers/world_responds/paper.md
R0067	reference note	Calibrated active learning literature	papers/probe_value_reengagement/paper.md

ID	Kind/status	Citation or reference item	Source files
R0068	unclear reference	BALD / value of information	papers/probe_value_reengagement/paper.md
R0069	unclear reference	Brehmer et al. (weakly supervised causal repr learning)	papers/probe_value_reengagement/paper.md
R0070	unclear reference	Empowerment (Klyubin et al.)	papers/probe_value_reengagement/paper.md
R0071	unclear reference	Di Paolo (autopoiesis/adaptivity)	papers/probe_value_reengagement/paper.md
R0072	internal repo	Paper 18 - papers/online_identifying_interventions/paper.md	papers/probe_value_reengagement/paper.md; papers/scale_normalize_d_vprobe/paper.md; papers/world_responds/paper.md
R0073	unclear reference	Carry over P22's stack plus:	papers/probe_value_reengagement/preregistration.md
R0074	unclear reference	Novelty detection / change-point detection: CUSUM, page-Hinkley, surprise-driven exploration	papers/probe_value_reengagement/preregistration.md
R0075	reference note	Active learning under distribution shift: continual active learning literature	papers/probe_value_reengagement/preregistration.md
R0076	reference note	Sense of agency under environmental drift: literature on agency maintenance through changing sensorimotor contingencies	papers/probe_value_reengagement/preregistration.md
R0077	reference note	Value of information in active inference: explicit treatment of intervention value vs information value Honest framing line: > Paper 23A redefines the program's probe-value signal as marginal MAE reducibility - the expected benefit of one more identifying intervention - and introduces a re-engagement mechanism that detects boundary staleness via non-null prediction surprise. Both upgrades address concrete bottlenecks Paper 22 named: oracle confounding (current error != value) and self-silencing (probe stops -> no data -> probe stays stopped).	papers/probe_value_reengagement/preregistration.md
R0078	unclear reference	Carried from P22-24's six-cluster citation stack. Add for P25:	papers/role_specific_identifiability/paper.md
R0079	unclear reference	Mixture of experts (Shazeer et al.) - disjoint per-role heads via gated routing	papers/role_specific_identifiability/paper.md
R0080	unclear reference	Causal abstraction identifiability under partial intervention (Beckers, Halpern)	papers/role_specific_identifiability/paper.md
R0081	unclear reference	Vervaeke relevance realization (lecture series) - relevance under representational bounds Plus the philosophical correlates the program operationalizes: Heidegger, Gibson, Uexkll, enactivism (Maturana/Varela/Thompson/Di Paolo), cybernetics (Ashby), active inference (Friston), pragmatism (Dewey), Canguilhem, Jonas, Simondon.	papers/role_specific_identifiability/paper.md
R0082	reference note	Per the program's framing literature stack (cited tight, 12-16 per paper). Calibrated active learning (closest external framing):	papers/scale_normalized_vprobe/paper.md
R0083	unclear reference	Calibrated Uncertainty Sampling for Active Learning	papers/scale_normalized_vprobe/paper.md; papers/world_responds/paper.md
R0084	unclear reference	When Active Learning Fails, Uncalibrated Out-of-Distribution Uncertainty Epistemic uncertainty calibration:	papers/scale_normalized_vprobe/paper.md
R0085	unclear reference	Epistemic Neural Networks	papers/scale_normalized_vprobe/paper.md; papers/world_responds/paper.md
R0086	unclear reference	Quantifying Epistemic Uncertainty in Deep Learning BALD / information gain / active inference epistemic value:	papers/scale_normalized_vprobe/paper.md
R0087	unclear reference	Bayesian Active Learning by Disagreements: A Geometric Perspective	papers/scale_normalized_vprobe/paper.md
R0088	unclear reference	Active Inference and Epistemic Value in Graphical Models Causal representation identifiability from interventions:	papers/scale_normalized_vprobe/paper.md
R0089	unclear reference	Brehmer et al., Weakly Supervised Causal Representation Learning	papers/scale_normalized_vprobe/paper.md; papers/world_responds/paper.md
R0090	unclear reference	General Identifiability and Achievability for CRL	papers/scale_normalized_vprobe/paper.md
R0091	unclear reference	On the Identifiability of Causal Abstractions Sense of agency / self-other attribution:	papers/scale_normalized_vprobe/paper.md
R0092	reference note	Comparator model literature	papers/scale_normalized_vprobe/paper.md
R0093	unclear reference	Predictive Processing Model of Perception and Action for Self-Other Distinction Homeostatic active inference + enactive adaptivity:	papers/scale_normalized_vprobe/paper.md

ID	Kind/status	Citation or reference item	Source files
R0094	unclear reference	Simulating homeostatic, allostatic and goal-directed forms of interoceptive control using Active Inference	papers/scale_normalized_vprobe/paper.md
R0095	unclear reference	Di Paolo, Autopoiesis, adaptivity, teleology, agency	papers/scale_normalized_vprobe/paper.md; papers/world_responds/paper.md
R0096	unclear reference	Empowerment: Universal Agent-Centric Measure of Control Foundations:	papers/scale_normalized_vprobe/paper.md
R0097	external bibliographic	Bennett (2023). On the computation of meaning, language models, and incomprehensible horrors.	papers/scale_normalized_vprobe/paper.md
R0098	external bibliographic	Levin (2022). Technological Approach to Mind Everywhere.	papers/scale_normalized_vprobe/paper.md
R0099	external bibliographic	Vervaeke (2019). Awakening from the Meaning Crisis.	papers/scale_normalized_vprobe/paper.md
R0100	external bibliographic	Locatello et al. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations.	papers/scale_normalized_vprobe/paper.md
R0101	internal repo	Paper 16b - papers/null_intervention/paper.md	papers/scale_normalized_vprobe/paper.md; papers/world_responds/paper.md
R0102	internal repo	Paper 17A - papers/costly_null_probes/paper.md	papers/scale_normalized_vprobe/paper.md; papers/world_responds/paper.md
R0103	unclear reference	Cite tight, ~12-16 papers across:	papers/scale_normalized_vprobe/preregistration.md
R0104	unclear reference	Calibrated active learning failures (closest external framing): Calibrated Uncertainty Sampling for Active Learning; When Active Learning Fails, Uncalibrated OOD Uncertainty.	papers/scale_normalized_vprobe/preregistration.md
R0105	unclear reference	Epistemic uncertainty calibration: Epistemic Neural Networks; Quantifying Epistemic Uncertainty in Deep Learning.	papers/scale_normalized_vprobe/preregistration.md
R0106	unclear reference	BALD / information gain / active inference: Bayesian Active Learning by Disagreements; Active Inference and Epistemic Value in Graphical Models.	papers/scale_normalized_vprobe/preregistration.md
R0107	unclear reference	Causal representation identifiability: Brehmer et al. Weakly Supervised Causal Representation Learning; General Identifiability and Achievability for CRL; Identifiability of Causal Abstractions.	papers/scale_normalized_vprobe/preregistration.md
R0108	reference note	Sense of agency: Comparator model literature; Predictive Processing Model for Self-Other Distinction.	papers/scale_normalized_vprobe/preregistration.md
R0109	unclear reference	Homeostatic active inference + Di Paolo + empowerment: Simulating homeostatic/allostatic active inference; Di Paolo on autopoiesis-adaptivity; Empowerment. Framing line (verbatim-usable): > We treat null actions as identifying interventions: costly epistemic actions whose value lies not in immediate reward but in reducing uncertainty about which future viability changes are self-caused versus world-caused. Prior work in active learning predicts uncertainty-driven acquisition works when uncertainty is calibrated, and fails when it is not. We show this failure mode inside a minimal self/world attribution problem and test a scale-normalized current-error correction. Claim discipline if 21A succeeds: > In a minimal two-variable homeostatic bandit, autonomous identifying interventions become vector-sensitive when probe uncertainty is calibrated in dimension-normalized units; without scale normalization, the agent's decision about when to identify is biased toward the larger-magnitude dimension. NOT: "we solved uncertainty" / "consciousness" / "agency solved."	papers/scale_normalized_vprobe/preregistration.md
R0110	internal repo	Paper 19 - Current-Error Calibration - papers/current_error_calibration/paper.md	papers/vector_first_order_self/paper.md

ID	Kind/status	Citation or reference item	Source files
R0111	external linked	Aubin, J.-P. (1991). Viability Theory. Birkhauser. Brown, J. (2026). The Metric Stack of Concern: From Viability Prediction to Maintained Self/World Boundaries in Minimal Agents. Das, A., Ghugarkar, O., Bhat, V., & Aali, A. (2026). Compositional Neuro-Symbolic Reasoning. arXiv:2604.02434. Cooper, P., & Velasquez, A. (2026). Active Causal Experimentalist (ACE): Learning intervention strategies via direct preference optimization. arXiv: 2602.02451. Elsen, T., Metzen, J. H., & Hutter, F. (2019). Neural architecture search: A survey. Journal of Machine Learning Research, 20(55), 1-21. Lehman, J., & Stanley, K. O. (2011). Abandoning objectives: Evolution through the search for novelty alone. Evolutionary Computation, 19(2), 189-223. Kim, K. W. (2026). Latent State Design for World Models under Sufficiency Constraints. arXiv:2605.01694. Le Lidec, Q., Biza, O., Goudet, O., Balestrieri, R., & Lajoie, G. (2026). Causal-JEPA: Learning World Models through Object-Level Latent Interventions. arXiv:2602.11389. Sharma, A., Nguyen, L., Gupta, A., Wang, C., Adebayo, C., & Kowalski, J. (2025). Inducing Causal World Models in LLMs for Zero-Shot Physical Reasoning. arXiv:2507.19855. Mouret, J.-B., & Clune, J. (2015). Illuminating search spaces by mapping elites. arXiv:1504.04909. Revencu, B., Pajot, M., & Dehaene, S. (2026). Representations of geometric shapes have syntactic structure. Journal of Experimental Psychology: General, 155(4), 1081-1102. https://doi.org/10.1037/xge0001890 Wu, Y.-F., Lee, M., & Ahn, S. (2024). Neural Language of Thought Models. arXiv:2402.01203. Yang, J., Zhang, D., Song, X., Dai, Q., Liu, X., Chen, Y., Vashishtha, A., Shi, J., Tan, C., & Peng, H. (2026). CausaLab: A scalable environment for interactive causal discovery toward AI scientists. arXiv:2605.26029. Zhang, S. et al. (2026). Structuring Open-Ended NAS: Semi-Automated Design Knowledge Structuring with LLMs for Efficient Neural Architecture Search. arXiv:2605.19247.	papers/viable_computational_bodies/paper.md
R0112	internal repo	papers/architecture_laws_machine_agency/paper.md	papers/virtual_governor_stress_signal/paper.md
R0113	internal repo	papers/causally_grounded_agents_benchmark/paper.md	papers/virtual_governor_stress_signal/paper.md
R0114	external linked	Bennett, M. T. How to Create Conscious Machines. arXiv:2403.00644 (2024).	papers/weakness_invariance_neurips/paper.md
R0115	external linked	Bennett, M. T. Are Flat Minima an Illusion? arXiv:2605.05209 (2026).	papers/weakness_invariance_neurips/paper.md
R0116	external linked	Bronstein, M. M., Bruna, J., Cohen, T., and Velickovic, P. Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges. arXiv:2104.13478 (2021).	papers/weakness_invariance_neurips/paper.md
R0117	external bibliographic	Cohen, T. and Welling, M. Group Equivariant Convolutional Networks. ICML (2016).	papers/weakness_invariance_neurips/paper.md
R0118	external bibliographic	D'Amour, A. et al. Underspecification Presents Challenges for Credibility in Modern Machine Learning. JMLR 23(226):1-61 (2022).	papers/weakness_invariance_neurips/paper.md
R0119	external bibliographic	Dinh, L., Pascanu, R., Bengio, S., and Bengio, Y. Sharp Minima Can Generalize for Deep Nets. ICML (2017).	papers/weakness_invariance_neurips/paper.md
R0120	external bibliographic	Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., and Wichmann, F. A. Shortcut Learning in Deep Neural Networks. Nature Machine Intelligence 2:665-673 (2020).	papers/weakness_invariance_neurips/paper.md
R0121	external bibliographic	Hochreiter, S. and Schmidhuber, J. Flat Minima. Neural Computation 9(1):1-42 (1997).	papers/weakness_invariance_neurips/paper.md
R0122	external bibliographic	Hutter, M. Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability. Springer (2005).	papers/weakness_invariance_neurips/paper.md
R0123	external bibliographic	Keskar, N. S., Mudigere, D., Nocedal, J., Smelyanskiy, M., and Tang, P. T. P. On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima. ICLR (2017).	papers/weakness_invariance_neurips/paper.md
R0124	external bibliographic	Kondor, R. and Trivedi, S. On the Generalization of Equivariance and Convolution in Neural Networks to the Action of Compact Groups. ICML (2018).	papers/weakness_invariance_neurips/paper.md
R0125	external bibliographic	Liu, Z., Michaud, E. J., and Tegmark, M. Omnigrok: Grokking Beyond Algorithmic Data. ICLR (2023).	papers/weakness_invariance_neurips/paper.md
R0126	external bibliographic	Mu, J. and Viswanath, P. All-but-the-Top: Simple and Effective Postprocessing for Word Representations. ICLR (2018).	papers/weakness_invariance_neurips/paper.md
R0127	external linked	Perin, A. and Deny, S. On the Ability of Deep Networks to Learn Symmetries from Data: A Neural Kernel Theory. arXiv:2412.11521 (2024).	papers/weakness_invariance_neurips/paper.md
R0128	external bibliographic	Power, A., Burda, Y., Edwards, H., Babuschkin, I., and Misra, V. Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets. ICLR Workshop (2022).	papers/weakness_invariance_neurips/paper.md
R0129	external bibliographic	Solomonoff, R. J. A Formal Theory of Inductive Inference, Parts I and II. Information and Control 7(1):1-22, 7(2):224-254 (1964).	papers/weakness_invariance_neurips/paper.md
R0130	external bibliographic	Valle-Perez, G., Camargo, C. Q., and Louis, A. A. Deep Learning Generalizes Because the Parameter-Function Map is Biased Towards Simple Functions. ICLR (2019).	papers/weakness_invariance_neurips/paper.md
R0131	external bibliographic	van der Ouderaa, T. F. A., Immer, A., and van der Wilk, M. Learning Layer-wise Equivariances Automatically using Gradients. NeurIPS (2023).	papers/weakness_invariance_neurips/paper.md

ID	Kind/status	Citation or reference item	Source files
R0132	external linked	Arjovsky, M., Bottou, L., Gulrajani, I., and Lopez-Paz, D. Invariant Risk Minimization. arXiv:1907.02893 (2019).	papers/weakness_invariance_neurips/paper.md
R0133	external linked	Zhou, K., Liu, Z., Qiao, Y., Xiang, T., and Loy, C. C. Domain Generalization: A Survey. arXiv:2103.02503 (2021).	papers/weakness_invariance_neurips/paper.md
R0134	external bibliographic	Rissanen, J. Modeling by Shortest Data Description. Automatica 14(5):465-471 (1978).	papers/weakness_invariance_neurips/paper.md
R0135	external bibliographic	Grunwald, P. D. The Minimum Description Length Principle. MIT Press (2007).	papers/weakness_invariance_neurips/paper.md
R0136	external linked	Mingard, C., Skalse, J., Valle-Perez, G., Martinez-Rubio, D., Mikulik, V., and Louis, A. A. Neural Networks are a Priori Biased Towards Boolean Functions with Low Entropy. arXiv:1909.11522 (2019).	papers/weakness_invariance_neurips/paper.md
R0137	reference note	Same six-cluster citation stack as P21A: Action-conditioned predictive models (new for P22): Ha & Schmidhuber world models; Dreamer family - explicit treatment of action-conditioned hidden state evolution. Causal mediator identification (new for P22): Pearl mediation analysis; relevance to direct vs mediated vs exogenous decomposition. Sense of agency under delayed/mediated consequences (new for P22): cognitive neuroscience on agency attribution when sensory feedback is delayed or mediated. Plus the established stack:	papers/world_responds/paper.md
R0138	unclear reference	Bennett, On the computation of meaning	papers/world_responds/paper.md
R0139	unclear reference	Levin, Technological Approach to Mind Everywhere	papers/world_responds/paper.md
R0140	unclear reference	Vervaeke, Awakening from the Meaning Crisis	papers/world_responds/paper.md
R0141	unclear reference	Active Inference and Epistemic Value in Graphical Models	papers/world_responds/paper.md
R0142	unclear reference	Empowerment (Universal Agent-Centric Measure of Control)	papers/world_responds/paper.md
R0143	internal repo	Paper 14b - papers/ensemble_uncertainty/paper.md	papers/world_responds/paper.md
R0144	internal repo	Paper 15 - papers/valence_tapestry/paper.md	papers/world_responds/paper.md
R0145	unclear reference	Yao, Shunyu and Zhao, Jeffrey and Yu, Dian and Du, Nan and Shafran, Izhak and Narasimhan, Karthik and Cao, Yuan (). ReAct: Synergizing Reasoning and Acting in Language Models. International Conference on Learning Representations	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0146	unclear reference	Shinn, Noah and Cassano, Federico and Berman, Edward and Gopinath, Ashwin and Narasimhan, Karthik and Yao, Shunyu (). Reflexion: Language Agents with Verbal Reinforcement Learning. Advances in Neural Information Processing Systems	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0147	unclear reference	Wang, Guanzhi and Xie, Yuqi and Jiang, Yunfan and Mandlekar, Ajay and Xiao, Chaowei and Zhu, Yuke and Fan, Linxi and Anandkumar, Anima (). Voyager: An Open-Ended Embodied Agent with Large Language Models. Transactions on Machine Learning Research	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0148	unclear reference	Ahn, Michael and Brohan, Anthony and Brown, Noah and Chebotar, Yevgen and Cortes, Omar and David, Byron and Finn, Chelsea and others (). Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. Conference on Robot Learning	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0149	unclear reference	Yao, Shunyu and Yu, Dian and Zhao, Jeffrey and Shafran, Izhak and Griffiths, Thomas L. and Cao, Yuan and Narasimhan, Karthik (). Tree of Thoughts: Deliberate Problem Solving with Large Language Models. Advances in Neural Information Processing Systems	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0150	unclear reference	Schick, Timo and Dwivedi-Yu, Jane and Dessi, Roberto and Raileanu, Roberta and Lomeli, Maria and Hambro, Eric and Zettlemoyer, Luke and Cancedda, Nicola and Scialom, Thomas (). Toolformer: Language Models Can Teach Themselves to Use Tools. Advances in Neural Information Processing Systems	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib
R0151	external linked	Liu, Xiao and Yu, Hao and Zhang, Hanchen and Xu, Yifan and Lei, Xuanyu and Lai, Hanyu and Gu, Yu and Ding, Hangliang and Men, Kaiwen and Yang, Kejuan and others (). AgentBench: Evaluating LLMs as Agents. arXiv preprint arXiv:2308.03688	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib

ID	Kind/status	Citation or reference item	Source files
R0152	unclear reference	Zhou, Shuyan and Xu, Frank F. and Zhu, Hao and Zhou, Xuhui and Lo, Robert and Sridhar, Abishek and Cheng, Xianyi and Bisk, Yonatan and Fried, Daniel and Alon, Uri and others (). WebArena: A Realistic Web Environment for Building Autonomous Agents. International Conference on Learning Representations	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0153	external linked	Xie, Tianbao and Zhang, Danyang and Chen, Jixuan and Li, Xiaochuan and Zhao, Siheng and Cao, Ruisheng and Hua, Toh and Cheng, Zhoujun and Shin, Dongchan and Lei, Fangyu and others (). OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. arXiv preprint arXiv:2404.07972	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0154	unclear reference	Mialon, Gregoire and Fourier, Clementine and Swift, Craig and Wolf, Thomas and LeCun, Yann and Scialom, Thomas (). GAIA: A Benchmark for General AI Assistants. International Conference on Learning Representations	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib
R0155	unclear reference	Jimenez, Carlos E. and Yang, John and Wettig, Alexander and Yao, Shunyu and Pei, Kexin and Press, Ofir and Narasimhan, Karthik (). SWE-bench: Can Language Models Resolve Real-World GitHub Issues?. International Conference on Learning Representations	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0156	unclear reference	Srivastava, Aarohi and Rastogi, Abhinav and Rao, Abhishek and Shueb, Abu Awal Md and others (). Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models. Transactions on Machine Learning Research	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib
R0157	external linked	Liang, Percy and Bommasani, Rishi and Lee, Tony and Tsipras, Dimitris and Soylu, Dilara and Yasunaga, Michihiro and others (). Holistic Evaluation of Language Models. arXiv preprint arXiv:2211.09110	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib
R0158	external linked	Amodei, Dario and Olah, Chris and Steinhardt, Jacob and Christiano, Paul and Schulman, John and Mane, Dan (). Concrete Problems in AI Safety. arXiv preprint arXiv:1606.06565	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0159	unclear reference	Langosco, Lauro and Koch, Jack and Sharkey, Lee and Pfau, Jacob and Orseau, Laurent and Krueger, David (). Goal Misgeneralization in Deep Reinforcement Learning. International Conference on Machine Learning	papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib; papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0160	unclear reference	D'Amour, Alexander and Heller, Katherine and Moldovan, Dan and Adlam, Ben and Alipanahi, Babak and Beutel, Alex and Chen, Christina and Deaton, Jonathan and Eisenstein, Jacob and Hoffman, Matthew D. and others (). Underspecification Presents Challenges for Credibility in Modern Machine Learning. Journal of Machine Learning Research	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0161	external linked	Arjovsky, Martin and Bottou, Leon and Gulrajani, Ishaan and Lopez-Paz, David (). Invariant Risk Minimization. arXiv preprint arXiv:1907.02893	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0162	unclear reference	Gulrajani, Ishaan and Lopez-Paz, David (). In Search of Lost Domain Generalization. International Conference on Learning Representations	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0163	unclear reference	Cohen, Taco and Welling, Max (). Group Equivariant Convolutional Networks. International Conference on Machine Learning	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0164	unclear reference	Sagawa, Shiori and Koh, Pang Wei and Hashimoto, Tatsunori B. and Liang, Percy (). Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization. International Conference on Learning Representations	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0165	unclear reference	Koh, Pang Wei and Sagawa, Shiori and Marklund, Henrik and Xie, Sang Michael and Zhang, Marvin and Balsubramani, Akshay and Hu, Weihua and Yasunaga, Michihiro and Phillips, Richard Lanase and Beery, Sara and others (). WILDS: A Benchmark of in-the-Wild Distribution Shifts. International Conference on Machine Learning	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0166	unclear reference	Peters, Jonas and Buehlmann, Peter and Meinshausen, Nicolai (). Causal Inference by Using Invariant Prediction: Identification and Confidence Intervals. Journal of the Royal Statistical Society: Series B	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0167	unclear reference	Geirhos, Robert and Jacobsen, Jorn-Henrik and Michaelis, Claudio and Zemel, Richard and Brendel, Wieland and Bethge, Matthias and Wichmann, Felix A. (). Shortcut Learning in Deep Neural Networks. Nature Machine Intelligence	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0168	unclear reference	Scholkopf, Bernhard and Locatello, Francesco and Bauer, Stefan and Ke, Nan Rosemary and Kalchbrenner, Nal and Goyal, Anirudh and Bengio, Yoshua (). Toward Causal Representation Learning. Proceedings of the IEEE	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib

ID	Kind/status	Citation or reference item	Source files
R0169	unclear reference	Locatello, Francesco and Bauer, Stefan and Lucic, Mario and Raetsch, Gunnar and Gelly, Sylvain and Schoelkopf, Bernhard and Bachem, Olivier (). Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations. International Conference on Machine Learning	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0170	unclear reference	Brehmer, Johann and De Haan, Pim and Lippe, Phillip and Cohen, Taco (). Weakly Supervised Causal Representation Learning. Advances in Neural Information Processing Systems	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0171	unclear reference	Hewitt, John and Liang, Percy (). Designing and Interpreting Probes with Control Tasks. EMNLP-IJCNLP	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0172	unclear reference	Meng, Kevin and Bau, David and Andonian, Alex and Belinkov, Yonatan (). Locating and Editing Factual Associations in GPT. Advances in Neural Information Processing Systems	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0173	unclear reference	Schick, Timo and Dwivedi-Yu, Jane and Dessì (). Toolformer: Language Models Can Teach Themselves to Use Tools. Advances in Neural Information Processing Systems	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0174	unclear reference	Khot, Tushar and Trivedi, Harsh and Finlayson, Matthew and Fu, Yao and Richardson, Kyle and Clark, Peter and Sabharwal, Ashish (). Decomposed Prompting: A Modular Approach for Solving Complex Tasks. International Conference on Learning Representations	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0175	unclear reference	Mialon, Grégoire and Fourier, Clémentine and Swift, Craig and Wolf, Thomas and LeCun, Yann and Scialom, Thomas (). GAIA: A Benchmark for General AI Assistants. International Conference on Learning Representations	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0176	unclear reference	Srivastava, Aarohi and Rastogi, Abhinav and Rao, Abhishek and Shueb, Abu Awal Md and Abid, Abubakar and Fisch, Adam and Brown, Adam R. and Santoro, Adam and Gupta, Aditya and Garriga-Alonso, Adrià and others (). Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models. Transactions on Machine Learning Research	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib
R0177	external linked	Liang, Percy and Bommasani, Rishi and Lee, Tony and Tsipras, Dimitris and Soylu, Dilara and Yasunaga, Michihiro and Zhang, Yian and Narayanan, Deepak and Wu, Yuhuai and Kumar, Ananya and others (). Holistic Evaluation of Language Models. arXiv preprint arXiv:2211.09110	papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib

Appendix B: Every Tracked PDF Reviewed

This appendix lists all 60 tracked PDFs reviewed by full-page text extraction.

PDF	Pages	Parsed title	Tags and summary
papers/allostatic_control/paper.pdf	10	Allostatic State Control	Tags: agents;geometry;concern;inquiry. Summary: Allostatic State Control Works at the Boundary - But Greedy Planning Beats Uncertainty-Aware Allostasis: An Honest Falsification of Four Pre-Registered Gates Author. Jawaun Brown. Abstract Companion paper [13b] showed that state-dependent valence is architecturally solvable (oracle boundary feature: return 50/50) but that smooth-function-approximator alternatives (Fourier features, sign-of-margin loss) reach state-con
papers/architecture_laws_machine_agency/paper.pdf	5	Architecture Laws for Machine Agency	Tags: causal;agents;geometry;concern;ood;inquiry. Summary: Architecture Laws for Machine Agency: A Synthesis of the Probe- Calibration Lineage Jawaun Brown 2026-07-06 Abstract This note distills the program's practical architecture lessons from the rst-order self, probe-calibration, world-responds, planning, and long-horizon papers. The empirical arc does not show consciousness, full agency, or general intelligence. It shows something narrower and more useful: small archite
papers/autopoietic_control/paper.pdf	9	From Active Geometry to Autopoietic Control	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: From Active Geometry to Autopoietic Control: Viability Slack as the Bottleneck for Adaptive Generalization Author. Jawaun Brown. Abstract Four prior empirical papers in this program established that symmetry-compatible-hypothesis volume (weakness) predicts out-of-distribution generalization [1], that the relevant symmetry group can be recovered from training data alone [2], that pixel-cosine and learned-encoder metho
papers/causally_grounded_agents_benchmark/paper.pdf	4	Benchmarking Causally Grounded Finite Agents	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Benchmarking Causally Grounded Finite Agents Condensed Design Note for a Proxy-Resistant Agent Benchmark Jawaun Brown 2026-07-06 Source Note This note is a benchmark-facing condensation of the research program in this repository. It is not a replacement for the full empirical reports. Portable source paths: umbrella benchmark: docs/causally_grounded_agents_benchmark.md release schema: docs/causally_grounded_agent
papers/comprehensive_literature_review/paper.pdf	15	From Concern to Action Surfaces	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: From Concern to Action Surfaces 1 From Concern to Action Surfaces A Critical Literature Review and Research Program for Causally Grounded Finite Agents Date: 2026-07-07 Abstract This paper reviews the research-derived experiments corpus as a unified program about finite agents: systems that must compress a world, preserve viability-relevant distinctions, act under uncertainty, and remain honest under distribution shi
papers/concern_bootstrap/paper.pdf	10	Bootstrapping Concern	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Bootstrapping Concern: DeltaE Auxiliary Loss Builds Valence Geometry on Easy Reward Structures, But Sparse-Reward Policies Cannot Exploit It Author. Jawaun Brown. Abstract Companion paper [7] left a clean open question: can valence-aligned representation self-organize from viability interaction, rather than being installed by supervised optimal-action pretraining? Three candidate mechanisms have been proposed: (a) an
papers/concerned_syntax/paper.pdf	18	Constituency Tests for Concerned Representation in Minimal Agents	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Constituency Tests for Concerned Representation in Minimal Agents Jawaun Brown 2026-06-22 Abstract Arc 1 of the maintained-concern program showed that a minimal homeostatic agent can detect boundary staleness, allocate costly null probes, saturate after identification, and re-engage after regime shifts. It also found a precise ceiling: under a shared-head, null-only intervention regime, better probe policy no longer c
papers/costly_null_probes/paper.pdf	10	Learning When Not to Act: Costly Null Probes for Self/World Identifiability in Minimal Homeostatic Agents	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Learning When Not to Act: Costly Null Probes for Self/ World Identifiability in Minimal Homeostatic Agents Jawaun Brown 2026-06-11 Abstract Paper 16b (Identifiability Through Intervention) showed that scheduled null actions, when used as world- only supervision, break the self/world gauge symmetry that defeats architectural factorization (Paper 16). The agent learned the rst-order self/world boundary through the act o
papers/current_error_calibration/paper.pdf	10	Current-Error Calibration for Identifying Interventions	Tags: causal;agents;geometry;concern;ood;inquiry. Summary: Current-Error Calibration for Identifying Interventions: Recent Residuals Are Not Enough Unless Recomputed Against the Present Model Jawaun Brown 2026-06-12 Abstract Paper 18 (Online Identifying Interventions) found that the debiased V_probe head - designed to x Paper 17A's noise- oor saturation - was instead anti-calibrated to oracle attribution uncertainty (Spearman = 0.55). The 5 diagnosis named "historical

PDF	Pages	Parsed title	Tags and summary
papers/ensemble_uncertainty/paper.pdf	8	When Models Don't Know What They Don't Know	Tags: agents;geometry;concern;ood;inquiry;consciousness. Summary: When Models Don't Know What They Don't Know: Bootstrap Ensembles Fail to Detect the Regime-Boundary Failure, Greedy Planning Remains Robust Author. Jawaun Brown. Abstract Companion paper [14] found that adding a regulate action lifts boundary-condition return from 24.5 to 42.5, but sophisticated planners (two-step lookahead, margin-condence) were worse than baseline because they used the model's own condence as a p
papers/epistemic_exploration/paper.pdf	9	Learning to Ask What Matters	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Learning to Ask What Matters: Conservative Epistemic Exploration Recovers Self-Organized Concern from a Biased Initial Policy; Novelty-Seeking Exploration Does Not Author. Jawaun Brown. Abstract Companion paper [10b] identified action coverage as a third necessary condition (alongside representational geometry and readout capacity) for self-organized concern-shaped competence: under a fully-biased "always consume" ini
papers/exploration_diagnostics/paper.pdf	8	Exploration Diagnostics	Tags: agents;geometry;concern;inquiry. Summary: Exploration Diagnostics: Skip-Branch Calibration Reveals Failure Modes; Margin-Based Epistemic Sampling Recovers Partially from Wrong-Init; All Mechanisms Fail Under High Noise Author. Jawaun Brown. Abstract Companion paper [11] reported that two intrinsic exploration mechanisms - greedy with decay and margin-based epistemic sampling (called expected_info_gain in [11]; here renamed to margin-based epistemic samplin
papers/first_order_self/paper.pdf	9	First-Order Self	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: First-Order Self: Architectural Factorization Alone Does Not Recover Self/World Attribution; Oracle Supervision Does - Gauge-Symmetric Heads Without Identifiability Author. Jawaun Brown. Abstract Companion paper [15] showed that vector DeltaV heads support exible concern under shifted internal-priority weights, while scalar drive heads cannot reweight. The natural next step - and the cleanest minimal computational te
papers/habituated_reengagement/paper.pdf	11	Habituated Re-Engagement: Post-Probe Cooling Stabilizes Autonomous Identifying Interventions	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Habituated Re-Engagement: Post-Probe Cooling Stabilizes Autonomous Identifying Interventions Jawaun Brown 2026-06-12 Abstract Paper 23A established that adding a non-null prediction-error boost to V_probe broke the self-silencing trap from Paper 22 - for the rst time in the program, the agent re-engaged probes after a regime shift (137% of pre-shift density, 3.04x of unaected buckets). But the same mechanism produc
papers/habituated_reengagement/suite_c_neural_probe_transfer.pdf	3	Suite C Neural Probe Transfer	Tags: agents;ood;inquiry;consciousness. Summary: Suite C Neural Probe Transfer: Learned Inquiry Without False Calm Jawaun Brown Abstract Suite C previously showed that a hand-specied decision-layer policy can re-engage after world change, recover attribution, quiet down without false calm, and reopen after a second shift. This paper tests the next bounded transfer: train the probe policy itself. A small NumPy MLP probe head is trained from Suite C teacher traces,
papers/habituated_reengagement/suite_c_reengagement_under_world_change.pdf	3	Suite C Re-Engagement Under World Change	Tags: causal;agents;ood;inquiry;consciousness. Summary: Suite C Re-Engagement Under World Change: A No-False-Calm Benchmark for Adaptive Inquiry Jawaun Brown Abstract Prior work in this lineage found a precise failure: agents can learn to probe eciently, become quiet after apparent convergence, and then fail to ask again after the world changes. Paper 23A restored re-engagement but produced anxious over-probing; Paper 23B showed that decision-layer cooling is safer than
papers/homeostatic_objects/paper.pdf	9	When Active Geometry Transfers	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: When Active Geometry Transfers: Valence-Pretrained Encoders Survive Episodic Homeostatic RL Where Sparse-Reward RL Fails Author. Jawaun Brown. Abstract The companion paper [6] showed that under a supervised optimal-action objective, encoders cluster the world by causal-valence role rather than sensory similarity - a clean static-objective demonstration of "objects from concern." This paper tests whether that demonstr
papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/paper.pdf	5	paper	Tags: causal;agents;geometry;ood;symmetry;inquiry. Summary: Agent benchmarks often reward final task success, but final success can be proxy-supported: a model may ex- ploit shortcuts, stale confidence, local final cues, uniform memory, or brittle tool surfaces. We propose a finite- agent benchmark design rule: a suite passes only when behavior, a suite-specific causal/structure gate, and anti- cheat controls all pass. We instantiate the rule with two hardened anchors. Suite

PDF	Pages	Parsed title	Tags and summary
papers/icml_publication_package_2026/structure_compatible_model_selection/paper.pdf	9	paper	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Structure-Compatible Model Selection Under Underspecification Anonymous Authors Abstract Modern learning pipelines are often underspecified: many trained models can match train and in-distribution validation metrics while behaving differently under deployment shift. We study a finite, auditable slice of this problem. The central hypothesis is that, when a deployment shift is generated by a candidate transformation
papers/icml_publication_package_2026/suite_c_teacher_free_inquiry/paper_outline.pdf	1	paper_outline	Tags: agents;ood;inquiry;consciousness. Summary: Suite C previously showed that a hand-specified policy and then a teacher-trained probe head can re-engage after world change. This paper reports the next finite step: a linear inquiry policy selected by teacher-free cross-entropy-method reward search. The training objective uses downstream recovery, selective re-engagement, second-shift reopenability, probe cost, and anti-cheat controls; it does not consume ha
papers/interventional_contrast/paper.pdf	10	Interventional Contrast for Mediated Self/World Attribution	Tags: causal;agents;geometry;ood;symmetry;inquiry. Summary: Interventional Contrast for Mediated Self/World Attribution: From Hand-Coded Buckets to Learned Probe Abstractions Jawaun Brown 2026-06-12 Abstract Paper 23B established the program's first stable maintained self/world boundary mechanism (8/10 gates, including G6 anti-cheating and G10 re-openability after a second regime shift). But Paper 23B's G8 (mediated/exogenous identifiability) added a remaining issue: the three
papers/learned_symmetry_discovery/paper.pdf	10	Learning the Group	Tags: causal;agents;ood;symmetry;inquiry;consciousness. Summary: Learning the Group: Data-Inferred Equivariance Predicts Out-of-Distribution Generalization Without Oracle Symmetry Author. Jawaun Brown. Abstract A central limitation of recent symmetry-and-generalization work - including our own prior result that weakness (symmetry-compatible-hypothesis volume) predicts OOD generalization on partially-observed cyclic symmetries - is that the transformation group is supplied by an or
papers/long_horizon_bottleneck/paper.pdf	8	Future Control Moves Memory	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Future Control Moves Memory: A Long-Horizon Moved-Bottleneck Diagnostic for Synthetic Agents Jawaun Brown 2026-07-06 Abstract This paper asks a narrow question about finite neural agents: when one early state variable becomes future-critical for a delayed decision, does the agent's memory geometry become specifically sensitive to that variable, and does that sensitivity move when the critical variable moves? We introduce
papers/metaphysics_synthesis/paper.pdf	14	paper	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: What Matters Becomes Measurable: A Synthesis of Structure-Compatible Generalization, Concern Geometry, and Proxy-Resistant Finite Agents Jawaun Brown Research Derived Experiments July 6, 2026 Abstract This synthesis condenses the July 2026 "Metaphysics of Intelligence" archive and the associated Research Derived Experiments repository into one reviewer-facing argument. The central result is not that the work proves c
papers/metric_stack_synthesis/paper.pdf	27	The Metric Stack of Concern	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: The Metric Stack of Concern: From Viability Prediction to Maintained Self/World Boundaries in Minimal Agents Jawaun Brown 2026-06-12 Abstract How can we tell whether a minimal agent's behavior depends on the intended internal structure, rather than on a proxy? We address this measurement problem in homeostatic bandits where agents must predict viability change, act, intervene through costly null actions, and attribute
papers/neural_group_generator/paper.pdf	8	When Pixels Beat Embeddings	Tags: causal;agents;geometry;ood;symmetry. Summary: When Pixels Beat Embeddings: Three Failed Neural Approaches to Symmetry Group Discovery Author. Jawaun Brown. Abstract The Learning the Group paper [2] showed that a simple enumerative procedure - for each candidate rotation in a 24-angle grid, score by mean pixel-cosine to the nearest same-class training example - recovers the true Z_8 rotation group from partial-orbit training data at 89.7% recall, 71.3% precision
papers/null_intervention/paper.pdf	10	Identifiability Through Intervention	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Identifiability Through Intervention: Null Actions Break the Gauge Symmetry of First-Order Self - Active Anchoring Reduces False-Credit by 82% Author. Jawaun Brown. Abstract Companion paper [16] established that architectural self/world factorization (self_head sees action; world_head doesn't) is gauge-symmetric: the two sub-heads can split the joint prediction arbitrarily, and architecture alone is insufficient for sem

PDF	Pages	Parsed title	Tags and summary
papers/off_policy_state_coverage/paper.pdf	9	Off-Policy State Coverage	Tags: causal;agents;geometry;concern;ood;inquiry. Summary: O-Policy DeltaE Training Partially Recovers State-Dependent Concern - But Refutes Paper 12's Coverage Diagnosis and Reveals a Boundary-Smoothing Failure at E=0.5 Author. Jawaun Brown. Abstract Companion paper [12] reported a uniform failure on state-dependent reward (return 13/50, action accuracy 0.47) across six online training conditions, including the Paper [11] winner (MBES) and the Paper [11b] ensemble-margin f
papers/online_identifying_interventions/paper.pdf	10	Online Identifying Interventions: Factorial Isolation of Probe-Target Bias and Data-Regime Effects in First-Order Self/World Identifiability	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Online Identifying Interventions: Factorial Isolation of Probe-Target Bias and Data-Regime Effects in First-Order Self/World Identifiability Jawaun Brown 2026-06-11 Abstract Paper 17A (Learning When Not to Act) falsified autonomous null-probe selection while replicating Paper 16b's anchor mechanism. Its 5 discussion named two distinct candidate bottlenecks: (i) the V_probe target was dominated by Bernoulli shock noise
papers/passive_to_active_geometry/paper.pdf	9	From Passive Cluster to Active Controller	Tags: causal;geometry;ood;symmetry;inquiry;consciousness. Summary: From Passive Cluster to Active Controller: Action Coupling Makes Latent Geometry Causally Load-Bearing Author. Jawaun Brown. Abstract Three prior empirical papers in this program established that weakness (symmetry-compatible-hypothesis volume) predicts out-of-distribution generalization [1], that the relevant symmetry group can often be recovered from training data alone [2], and that pixel-cosine and learned-encode
papers/pdf/concern_deforms_metric.pdf	12	Value-Weighted Training Deforms Learned Metrics Across RNN, Transformer, and JEPA-Style Spatial Models	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Value-Weighted Training Deforms Learned Metrics Across RNN, Transformer, and JEPA-Style Spatial Models Jawaun Brown Research-Derived Experiments Paper B Abstract If a representation is used to act, valuable regions should not merely receive labels; they should receive resolution. This paper tests a bounded version of that claim by injecting a movable loss-weight field, called concern here only as operational shorth
papers/pdf/reward_deformation_effective_dimension_law.pdf	6	A Measured Effective-Dimension Law for Value-Weighted Metric Deformation in Path-Integration RNNs	Tags: causal;agents;geometry;concern. Summary: A Measured Effective-Dimension Law for Value-Weighted Metric Deformation in Path-Integration RNNs Jawaun Brown Research-Derived Experiments Preregistered Geometry Sweep Abstract A value-weighted rate-distortion derivation predicts that a finite-capacity 2-D code should allocate local area density as $\det g^{-1/2}$. We preregistered a geometry sweep designed to distinguish this physical 2-D law from an effectively
papers/pdf/unified_metric_weakness_portfolio/finite_representations_portfolio.pdf	38	finite_representations_portfolio	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Finite Representations Allocate Resolution and Generalization Under Structure Four Research-Derived Experiments on Metric Deformation, Effective Dimension, Weakness, and Toroidal Topology Jawaun Brown July 2026 Executive Overview This portfolio collects four linked research-derived experiments on finite learned representations. 1. Metric allocation is experimentally movable. In finite-capacity spatial models, external
papers/pdf/unified_metric_weakness_portfolio/finite_representations_portfolio_with_bookmarks.pdf	38	finite_representations_portfolio_with_bookmarks	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Finite Representations Allocate Resolution and Generalization Under Structure Four Research-Derived Experiments on Metric Deformation, Effective Dimension, Weakness, and Toroidal Topology Jawaun Brown July 2026 Executive Overview This portfolio collects four linked research-derived experiments on finite learned representations. 1. Metric allocation is experimentally movable. In finite-capacity spatial models, external
papers/pdf/weakness_predicts_ood.pdf	6	Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization	Tags: causal;geometry;ood;symmetry;inquiry. Summary: Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization Jawaun Brown Weakness, not generic compression alone, in shortcut-compatible learning problems Research-Derived Experiments preprint compiled from the project repository Abstract We study finite-domain model selection where training data admits both a local shortcut and a globally symmetry-compatible rule. Loss and in-distribution val
papers/pdf/weakness_predicts_topology.pdf	7	Translation Augmentation Produces Toroidal Codes and Larger-Arena Generalization in Path-Integration RNNs	Tags: causal;agents;geometry;ood;symmetry;inquiry. Summary: Translation Augmentation Produces Toroidal Codes and Larger-Arena Generalization in Path-Integration RNNs Jawaun Brown Research-Derived Experiments empirical note and negative mediation result Abstract This note tests a pre-registered strong claim about path-integration RNNs: weakness, a held-out linear equivariance score for wrapped translations, should govern whether a learned population code forms a toroidal manifold

PDF	Pages	Parsed title	Tags and summary
papers/phase4_metaphysics/learning_missing_conditions.pdf	3	Learning the Missing Conditions	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Learning the Missing Conditions Jawaun Brown Phase 4 diagnostics for the Metric Stack of Concern Research-Derived Experiments - controlled L4-parallel harnesses Abstract Phase 3 identified several open bottlenecks in the Metric Stack of Concern: language scale, non-enumerative symmetry discovery, learned regime variables, probe value, architecture beyond the mediated-identifiability ceiling, foundation-model-style me
papers/phase5_external_validity/from_controlled_concern_geometry_to_foundation_models.pdf	3	From Controlled Concern Geometry to Foundation Models	Tags: causal;agents;geometry;concern;ood;memory. Summary: From Controlled Concern Geometry to Foundation Models Jawaun Brown Phase 5 external-validity transport for the Metric Stack of Concern Research-Derived Experiments - budgeted L4-parallel proxy suite Abstract Phase 4 showed that several missing conditions in the Metric Stack of Concern can be learned inside controlled diagnostic harnesses. Phase 5 asks which mechanisms transport when the setup becomes more model-like,
papers/phase6_real_model_validation/from_proxy_transport_to_open_models.pdf	2	From Proxy Transport to Open Models	Tags: geometry;concern. Summary: From Proxy Transport to Open Models Jawaun Brown Phase 6 actual-model validation for the Metric Stack of Concern Research-Derived Experiments - budgeted Modal L4 sweep Abstract Phase 5 used proxy harnesses to decide whether expensive validation was warranted. Phase 6 runs that heavier tier directly: public decoder language models are scored by hidden-state geometry and action logprob margins, while public frozen sent
papers/planning_from_concern/paper.pdf	11	Planning from Concern	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Planning from Concern: Model-Based DeltaE Action Selection Yields Self-Organized Homeostatic Competence Without Optimal-Action Supervision Author. Jawaun Brown. Abstract Companion paper [9] showed that an action-conditioned DeltaE auxiliary loss can self-organize a reward-aligned encoder representation (XOR cluster gap +1.84 without supervised optimal-action labels), but companion paper [8] showed that sparse-reward
papers/planning_hardening/paper.pdf	13	Distributed Concern	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Distributed Concern: Planning from DeltaE Depends on Latent Geometry, Not a Single Reward Axis Author. Jawaun Brown. Abstract Companion paper [10] reported that a DeltaE-aux-organized encoder + greedy argmax_a DeltaE_head(z, E, a) achieves return 50/50 and action accuracy 0.996 on the XOR homeostatic bandit, without supervised optimal-action labels and without policy-gradient training, and oered the natural-seeming
papers/probe_value_reengagement/paper.pdf	9	Probe Value and Re-Engagement: Learning When to Ask Again in Responsive Worlds	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Probe Value and Re-Engagement: Learning When to Ask Again in Responsive Worlds Jawaun Brown 2026-06-12 Abstract Paper 22 left two open issues. First, the program's oracle_probe_value condition used current attribution error as the signal, achieving nal lc_MAE 5x worse than learned probing - empirically falsifying it as an upper bound (current error != value of information). Second, the agent's probe completely stopp
papers/regime_sensitive_de/paper.pdf	8	Regime-Sensitive DeltaE Models	Tags: causal;agents;geometry;concern;ood. Summary: Regime-Sensitive DeltaE Models Solve State-Dependent Concern at the Diagnostic Limit: An Oracle Boundary Feature Closes the Gap; Learned Alternatives Solve Everywhere Except the Singular Point E=0.5 Author. Jawaun Brown. Abstract Companion paper [13a] showed that o-policy DeltaE training partially recovers state-dependent concern (action accuracy 0.96 on state_dep_inv_xor, vs the online failure oor 0.47) but leaves
papers/role_specific_identifiability/paper.pdf	9	Role-Specific Mediated Effects, Two-Sided Gauge Anchoring, and Fully-Learned Probe Abstractions	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Role-Specic Mediated Eects, Two-Sided Gauge Anchoring, and Fully-Learned Probe Abstractions: The Architectural Endpoint of the Autonomous-Probing Arc Jawaun Brown 2026-06-12 Abstract Paper 24 closed most of the mediated/exogenous identifiability gap from Paper 23B via interventional contrast loss (G2 56% mediated MAE reduction). But G7 caught a structural nding: wrong-history contrast pairs also helped, because t
papers/scale_normalized_vprobe/paper.pdf	12	Scale-Normalized Probe Calibration	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Scale-Normalized Probe Calibration for Vector First-Order Self: A Target x Threshold Factorial Jawaun Brown 2026-06-12 Abstract Paper 20B idntified a scale-asymmetric V_probe calibration failure in vector rst-order self: the E dimension calibrated partially (Spearman = +0.20), the D dimension was anti-calibrated (= 0.41), and learned probing performed 28% worse than matched-random at matched null count. The dia

PDF	Pages	Parsed title	Tags and summary
papers/state_dependent_concern/paper.pdf	8	State-Dependent Concern Fails	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: State-Dependent Concern Fails Under Online Homeostatic Training: A Coupling Between Policy and Training Distribution Author. Jawaun Brown. Abstract Companion paper [11] established that the model_plan_delta_e pipeline ([10]) can self-organize concern- shaped competence under various exploration regimes. Companion paper [11b] tightened that result with calibration metrics and identified margin-based epistemic sampling
papers/structure_compatible_generalization/inferred_transformations_intervention.pdf	2	Inferred Transformations for Structure-Compatible Generalization	Tags: causal;ood;symmetry. Summary: Inferred Transformations for Structure-Compatible Generalization Jawaun Brown Abstract Phase one showed that oracle transformation compatibility predicts OOD behavior under underspecification. This phase asks whether the oracle can be weakened: infer supported transformations from training evidence, score compatibility under that discovered family, and use the discovered family as a train-time regularizer. In a modula
papers/structure_compatible_generalization/language_template_substitution.pdf	3	Language-Template Substitution for Structure-Compatible Generalization	Tags: causal;concern;ood;symmetry. Summary: Language-Template Substitution for Structure-Compatible Generalization Jawaun Brown Phase 4: finite text-template generator transfer Abstract This phase tests structure-compatible generalization in a controlled finite text domain. Examples are rendered as short language templates with number words and offset words, while the deployment shift is precise: held-out number-word substitutions transport the label by the sa
papers/structure_compatible_generalization/learned_generators_transfer.pdf	3	Learned Generators for Structure-Compatible Generalization	Tags: causal;ood;symmetry. Summary: Learned Generators for Structure-Compatible Generalization Jawaun Brown Phase 3: finite generator discovery plus transfer Abstract Earlier structure-compatible generalization results used oracle deployment transformations or finite shifts inferred from observed overlaps. Phase three learns finite generator families from data: modular affine input/label transports and vision rotations selected by training-set self-con
papers/structure_compatible_generalization/semantic_retrieval_transfer.pdf	3	Semantic Retrieval Transfer for Structure-Compatible Generalization	Tags: ood. Summary: Semantic Retrieval Transfer for Structure-Compatible Generalization Jawaun Brown Phase 5: frozen-encoder semantic transfer Abstract This phase moves structure-compatible generalization from rendered templates to short semantic retrieval cases. Actual frozen sentence encoders provide embeddings; nearest-neighbor structure infers candidate paraphrase and entity-substitution orbits; trained retrieval selectors are evalu
papers/structure_compatible_generalization/semantic_selection_control.pdf	2	Semantic Selection Control for Structure-Compatible Generalization	Tags: concern;ood;symmetry. Summary: Semantic Selection Control for Structure-Compatible Generalization Jawaun Brown Phase 6: OOD-free frozen-encoder model selection Abstract This phase turns semantic retrieval compatibility into an OOD-free model-selection protocol. Candidate retrieval models are grouped into encoder-threshold seed zoos, filtered to high train and ID validation performance, selected by learned compatibility or baseline selectors, and e
papers/structure_compatible_generalization/structure_compatible_generalization.pdf	2	Structure-Compatible Generalization	Tags: agents;ood;symmetry. Summary: Structure-Compatible Generalization Jawaun Brown Abstract Modern learning systems are underspecified: many nite models can t the same training data and pass the same in- distribution checks while behaving dierently under deployment shifts. This paper reports a controlled diagnostic suite testing whether the transformation structure a learned function preserves predicts OOD behavior better than loss, ID validation,
papers/two_bottlenecks/paper.pdf	10	Two Bottlenecks	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Two Bottlenecks: DeltaE Auxiliary Bootstraps XOR Valence Geometry When Decoupled from Sparse-Reward Policy Gradient Author. Jawaun Brown. Abstract Companion paper [8] found a clean decoupling cell: an encoder trained with an action-conditioned DeltaE auxiliary loss achieved reward-axis cluster gap +1.00 on the additive_thresh reward function, while sparse-reward REINFORCE on the same encoder + policy combination stay
papers/valence_object_formation/paper.pdf	8	Objects Form from Concern	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Objects Form from Concern: Valence-Coupled Encoders Cluster the World by Causal Role, Not Sensory Similarity Author. Jawaun Brown. Abstract The ve prior empirical papers in this program established that weakness predicts OOD generalization [1], that the relevant symmetry group can be inferred from data [2], that pixel-cosine and learned-encoder methods occupy dierent operating regimes [3], that paraphrase clusters

PDF	Pages	Parsed title	Tags and summary
papers/valence_tapestry/paper.pdf	10	Tapestry of Valence	Tags: causal;agents;geometry;concern;ood;inquiry. Summary: Tapestry of Valence: Vector DeltaV Models Adapt Medicine Handling Under Internal-Weight Shifts; Scalar Drive Models Cannot Author. Jawaun Brown. Abstract Companion paper [14b] showed that bootstrap ensembles of identical-architecture DeltaE heads fail to detect regime-boundary failures, and Paper [14] showed that greedy planning + a safe regulate fallback action robustly beats sophisticated planners under model error
papers/vector_first_order_self/paper.pdf	12	Vector First-Order Self	Tags: causal;agents;geometry;concern;ood;symmetry. Summary: Vector First-Order Self: Autonomous Identifying Interventions for Multi-Valence Homeostatic Agents Subtitle: Per-dimension self/world attribution under autonomous current-replay null probing. Jawaun Brown 2026-06-12 Abstract The program's three strongest scalar mechanisms - Paper 15's vector DeltaV head, Paper 16b's null-anchor intervention, and Paper 19's current-replay autonomous probe - exist independently. Paper
papers/viable_computational_bodies/paper.pdf	16	Viability-Guided Evolution of Syntax-Bearing Computational Bodies	Tags: causal;agents;geometry;concern;symmetry;inquiry. Summary: Viability-Guided Evolution of Syntax-Bearing Computational Bodies Jawaun Brown 2026-06-22 Abstract Arc 2A asks what interventions make the world's causal grammar visible. Arc 2B asks what computational bodies can express that grammar in the rst place. We introduce a typed symbolic architecture-evolution benchmark for syntax-bearing agents. Candidate bodies are motif sets containing components such as tree binders, s
papers/virtual_governor_stress_signal/paper.pdf	3	Virtual-Governor Stress Signals for Local Action Recovery	Tags: causal;agents;ood;memory;consciousness. Summary: Virtual-Governor Stress Signals for Local Action Recovery Jawaun Brown Abstract The virtual-governor preprint is useful because it names a signal architecture: global constraint violations become local incentives. This paper turns that phrase into a bounded neural diagnostic. Small policies act in a three-dimensional controlled state whose target changes during long rollouts. Five architectures receive the same oracl
papers/weakness_invariance_neurips/paper.pdf	16	Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization	Tags: agents;geometry;ood;symmetry;consciousness. Summary: Symmetry-Compatible Hypothesis Volume Predicts Out- of-Distribution Generalization Subtitle. Weakness, not generic compression alone, in shortcut-compatible learning problems. Author. Jawaun Brown. Abstract We study model selection in nite-domain settings where the training data admits both a local shortcut and a globally symmetry-compatible rule. In these settings, training loss and in-distribution validation canno
papers/world_responds/paper.pdf	9	When the World Responds	Tags: causal;agents;geometry;concern;ood;inquiry. Summary: When the World Responds: Action-Correlated Shocks and the Limits of Null-Anchored Self/World Attribution Jawaun Brown 2026-06-12 Abstract Papers 16b through 21A established autonomous self/world attribution mechanisms for environments where world shocks are independent of the agent's prior actions. Paper 22 tests the rst regime where this independence breaks: a hidden hazard $h(t+1) = h(t) + I[\text{consume_trigger}]$ wh

Appendix C: Tracked Paper Source Inventory

File	Kind	Title
papers/allostatic_control/paper.md	primary_paper	Allostatic State Control Works at the Boundary - But Greedy Planning Beats Uncertainty-Aware Allostasis: An Honest Falsification of Four Pre-Registered Gates
papers/architecture_laws_machine_agency/paper.md	primary_paper	Architecture Laws for Machine Agency: A Synthesis of the Probe-Calibration Lineage
papers/autopoietic_control/paper.md	primary_paper	From Active Geometry to Autopoietic Control: Viability Slack as the Bottleneck for Adaptive Generalization
papers/causally_grounded_agents_benchmark/paper.md	primary_paper	Benchmarking Causally Grounded Finite Agents
papers/comprehensive_literature_review/paper.md	primary_paper	From Concern to Action Surfaces
papers/concern_bootstrap/paper.md	primary_paper	Bootstrapping Concern: DeltaE Auxiliary Loss Builds Valence Geometry on Easy Reward Structures, But Sparse-Reward Policies Cannot Exploit It
papers/concerned_syntax/paper.md	primary_paper	Constituency Tests for Concerned Representation in Minimal Agents
papers/concerned_syntax/preregistration.md	preregistration_or_runbook	Paper 2A Pre-Registration
papers/costly_null_probes/paper.md	primary_paper	Learning When Not to Act: Costly Null Probes for Self/World Identifiability in Minimal Homeostatic Agents
papers/costly_null_probes/preregistration.md	preregistration_or_runbook	Paper 17A - Pre-Registration
papers/current_error_calibration/paper.md	primary_paper	Current-Error Calibration for Identifying Interventions: Recent Residuals Are Not Enough Unless Recomputed Against the Present Model
papers/current_error_calibration/preregistration.md	preregistration_or_runbook	Paper 19 - Pre-Registration
papers/ensemble_uncertainty/paper.md	primary_paper	When Models Don't Know What They Don't Know: Bootstrap Ensembles Fail to Detect the Regime-Boundary Failure, Greedy Planning Remains Robust
papers/epistemic_exploration/paper.md	primary_paper	Learning to Ask What Matters: Conservative Epistemic Exploration Recovers Self-Organized Concern from a Biased Initial Policy; Novelty-Seeking Exploration Does Not
papers/exploration_diagnostics/paper.md	primary_paper	Exploration Diagnostics: Skip-Branch Calibration Reveals Failure Modes; Margin-Based Epistemic Sampling Recovers Partially from Wrong-Init; All Mechanisms Fail Under High Noise
papers/first_order_self/paper.md	primary_paper	First-Order Self: Architectural Factorization Alone Does Not Recover Self/World Attribution; Oracle Supervision Does - Gauge-Symmetric Heads Without Identifiability
papers/grid_cell_weakness/preregistration.md	preregistration_or_runbook	Paper A Scale-Up Pre-Registration
papers/grid_cell_weakness/runbook.md	preregistration_or_runbook	Paper A Runbook - Grid-Cell Weakness Sweep
papers/habituated_reengagement/paper.md	primary_paper	Habituated Re-Engagement: Post-Probe Cooling Stabilizes Autonomous Identifying Interventions
papers/habituated_reengagement/preregistration.md	preregistration_or_runbook	Paper 23B - Pre-Registration
papers/habituated_reengagement/suite_c_neural_probe_transfer.md	supporting_paper_source	Suite C Neural Probe Transfer: Learned Inquiry Without False Calm
papers/habituated_reengagement/suite_c_reengagement_under_world_change.md	supporting_paper_source	Suite C Re-Engagement Under World Change: A No-False-Calm Benchmark for Adaptive Inquiry
papers/homeostatic_objects/paper.md	primary_paper	When Active Geometry Transfers: Valence-Pretrained Encoders Survive Episodic Homeostatic RL Where Sparse-Reward RL Fails
papers/icml_publication_package_2026/README.md	readme	ICML Publication Package 2026
papers/icml_publication_package_2026/metric_stack_position_package/README.md	readme	Metric Stack / Concern Geometry Position Package
papers/icml_publication_package_2026/metric_stack_position_package/architecture_laws.md	supporting_paper_source	Architecture Laws for Concern-Mediated Finite Agents
papers/icml_publication_package_2026/metric_stack_position_package/position_outline.md	supporting_paper_source	Position Paper Outline: The Geometry of Concern
papers/icml_publication_package_2026/metric_stack_position_package/publication_sequence.md	supporting_paper_source	Publication Sequence
papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/README.md	readme	ICML/Benchmark-Style Package: Proxy-Resistant Finite-Agent Benchmarks

File	Kind	Title
papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/appendix.tex	supporting_paper_source	appendix
papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/figures_manifest.md	supporting_paper_source	Figure Manifest
papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/paper.tex	primary_paper	Proxy-Resistant Benchmarks for Causally Grounded Finite Agents
papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/references.bib	bibliography	references
papers/icml_publication_package_2026/proxy_resistant_agent_benchmarks/release_checklist.md	supporting_paper_source	Release Checklist
papers/icml_publication_package_2026/structure_compatible_model_selection/README.md	readme	ICML-Style Paper Package: Structure-Compatible Model Selection
papers/icml_publication_package_2026/structure_compatible_model_selection/appendix.tex	supporting_paper_source	appendix
papers/icml_publication_package_2026/structure_compatible_model_selection/figures_manifest.md	supporting_paper_source	Figure Manifest
papers/icml_publication_package_2026/structure_compatible_model_selection/paper.tex	primary_paper	Structure-Compatible Model Selection Under Underspecification
papers/icml_publication_package_2026/structure_compatible_model_selection/references.bib	bibliography	references
papers/icml_publication_package_2026/structure_compatible_model_selection/result_tables.md	supporting_paper_source	Result Tables for the ICML-Style Package
papers/icml_publication_package_2026/structure_compatible_model_selection/reviewer_risk_audit.md	supporting_paper_source	Reviewer Risk Audit
papers/icml_publication_package_2026/structure_compatible_model_selection/statistical_analysis_plan.md	supporting_paper_source	Statistical Analysis Plan
papers/icml_publication_package_2026/structure_compatible_model_selection/submission_polish_checklist.md	supporting_paper_source	Submission Polish Checklist
papers/icml_publication_package_2026/structure_compatible_model_selection/validation_checklist.md	supporting_paper_source	Validation Checklist
papers/icml_publication_package_2026/suite_c_teacher_free_inquiry/README.md	readme	Suite C Teacher-Free Inquiry Package
papers/icml_publication_package_2026/suite_c_teacher_free_inquiry/experiment_design.md	supporting_paper_source	Experiment Design: Teacher-Free Suite C Inquiry
papers/icml_publication_package_2026/suite_c_teacher_free_inquiry/paper_outline.tex	supporting_paper_source	Teacher-Free Adaptive Inquiry Under World Change
papers/icml_publication_package_2026/suite_c_teacher_free_inquiry/preregistration.md	preregistration_or_runbook	Preregistration: Suite C Teacher-Free Inquiry
papers/interventional_contrast/paper.md	primary_paper	Interventional Contrast for Mediated Self/World Attribution: From Hand-Coded Buckets to Learned Probe Abstractions
papers/interventional_contrast/preregistration.md	preregistration_or_runbook	Paper 24 - Pre-Registration
papers/learned_symmetry_discovery/paper.md	primary_paper	Learning the Group: Data-Inferred Equivariance Predicts Out-of-Distribution Generalization Without Oracle Symmetry
papers/long_horizon_bottleneck/api_blackbox_preregistration.md	preregistration_or_runbook	API Black-Box Behavioral Benchmark Preregistration
papers/long_horizon_bottleneck/autoregressive_json_surface_preregistration.md	preregistration_or_runbook	Autoregressive JSON Surface Transition Record
papers/long_horizon_bottleneck/generated_json_surface_preregistration.md	preregistration_or_runbook	Generated JSON Surface Transition Record
papers/long_horizon_bottleneck/paper.md	primary_paper	Future Control Moves Memory: A Long-Horizon Moved-Bottleneck Diagnostic for Synthetic Agents
papers/long_horizon_bottleneck/preregistration.md	preregistration_or_runbook	Preregistration: Long-Horizon Moved Bottleneck
papers/long_horizon_bottleneck/prompt_json_causal_patch_preregistration.md	preregistration_or_runbook	Prompt JSON Fixed-Prefix Causal Patch Preregistration
papers/long_horizon_bottleneck/prompt_json_fixed_action_localization_preregistration.md	preregistration_or_runbook	Prompt JSON Fixed-Action Localization Preregistration
papers/long_horizon_bottleneck/prompt_json_hidden_localization_preregistration.md	preregistration_or_runbook	Prompt JSON Hidden-Localization Preregistration

File	Kind	Title
papers/long_horizon_bottleneck/prompt_json_prompt_family_causal_patch_preregistration.md	preregistration_or_runbook	Prompt JSON Prompt-Family Causal Patch Preregistration
papers/long_horizon_bottleneck/prompt_json_transfer_preregistration.md	preregistration_or_runbook	Prompt JSON Transfer Preregistration
papers/long_horizon_bottleneck/text_argument_surface_preregistration.md	preregistration_or_runbook	Text Argument-Surface Transition Record
papers/metaphysics_synthesis/README.md	readme	Metaphysics of Intelligence Synthesis
papers/metaphysics_synthesis/paper.tex	primary_paper	What Matters Becomes Measurable: A Synthesis of Structure-Compatible Generalization, Concern Geometry, and Proxy-Resistant Finite Agents
papers/metric_stack_synthesis/paper.md	primary_paper	The Metric Stack of Concern: From Viability Prediction to Maintained Self/World Boundaries in Minimal Agents
papers/neural_group_generator/paper.md	primary_paper	When Pixels Beat Embeddings: Three Failed Neural Approaches to Symmetry Group Discovery
papers/null_intervention/paper.md	primary_paper	Identifiability Through Intervention: Null Actions Break the Gauge Symmetry of First-Order Self - Active Anchoring Reduces False-Credit by 82%
papers/off_policy_state_coverage/paper.md	primary_paper	Off-Policy DeltaE Training Partially Recovers State-Dependent Concern - But Refutes Paper 12's Coverage Diagnosis and Reveals a Boundary-Smoothing Failure at E=0.5
papers/online_identifying_interventions/paper.md	primary_paper	Online Identifying Interventions: Factorial Isolation of Probe-Target Bias and Data-Regime Effects in First-Order Self/World Identifiability
papers/online_identifying_interventions/preregistration.md	preregistration_or_runbook	Paper 18 - Pre-Registration
papers/passive_to_active_geometry/paper.md	primary_paper	From Passive Cluster to Active Controller: Action Coupling Makes Latent Geometry Causally Load-Bearing
papers/pdf/README.md	readme	Compiled Paper PDFs
papers/phase4_metaphysics/paper.md	primary_paper	Learning the Missing Conditions
papers/phase5_external_validity/paper.md	primary_paper	From Controlled Concern Geometry to Foundation Models
papers/phase6_real_model_validation/paper.md	primary_paper	From Proxy Transport to Open Models
papers/planning_from_concern/paper.md	primary_paper	Planning from Concern: Model-Based DeltaE Action Selection Yields Self-Organized Homeostatic Competence Without Optimal-Action Supervision
papers/planning_hardening/paper.md	primary_paper	Distributed Concern: Planning from DeltaE Depends on Latent Geometry, Not a Single Reward Axis
papers/probe_value_reengagement/paper.md	primary_paper	Probe Value and Re-Engagement: Learning When to Ask Again in Responsive Worlds
papers/probe_value_reengagement/preregistration.md	preregistration_or_runbook	Paper 23A - Pre-Registration
papers/regime_sensitive_de/paper.md	primary_paper	Regime-Sensitive DeltaE Models Solve State-Dependent Concern at the Diagnostic Limit: An Oracle Boundary Feature Closes the Gap; Learned Alternatives Solve Everywhere Except the Singular Point E=0.5
papers/role_specific_identifiability/paper.md	primary_paper	Role-Specific Mediated Effects, Two-Sided Gauge Anchoring, and Fully-Learned Probe Abstractions: The Architectural Endpoint of the Autonomous-Probing Arc
papers/role_specific_identifiability/preregistration.md	preregistration_or_runbook	Paper 25 - Pre-Registration
papers/scale_normalized_vprobe/paper.md	primary_paper	Scale-Normalized Probe Calibration for Vector First-Order Self: A Target x Threshold Factorial
papers/scale_normalized_vprobe/preregistration.md	preregistration_or_runbook	Paper 21A - Pre-Registration
papers/semantic_concern_geometry/preregistration.md	preregistration_or_runbook	Pre-registration: Semantic Concern Geometry
papers/state_dependent_concern/paper.md	primary_paper	State-Dependent Concern Fails Under Online Homeostatic Training: A Coupling Between Policy and Training Distribution
papers/structure_compatible_generalization/README.md	readme	Structure-Compatible Generalization Paper
papers/structure_compatible_generalization/inferred_transformations_intervention.md	supporting_paper_source	Inferred Transformations for Structure-Compatible Generalization
papers/structure_compatible_generalization/language_template_substitution.md	supporting_paper_source	Language-Template Substitution for Structure-Compatible Generalization
papers/structure_compatible_generalization/language_template_substitution_preregistration.md	preregistration_or_runbook	Phase 4 Preregistration: Language-Template Substitution for SCG
papers/structure_compatible_generalization/learned_generators_transfer.md	supporting_paper_source	Learned Generators for Structure-Compatible Generalization

File	Kind	Title
papers/structure_compatible_generalization/phase3_learned_generators_preregistration.md	preregistration_or_runbook	Phase 3 Preregistration: Learned Generators for Structure-Compatible Generalization
papers/structure_compatible_generalization/semantic_retrieval_transfer.md	supporting_paper_source	Semantic Retrieval Transfer for Structure-Compatible Generalization
papers/structure_compatible_generalization/semantic_retrieval_transfer_preregistration.md	preregistration_or_runbook	Phase 5 Preregistration: Semantic Retrieval Transfer for SCG
papers/structure_compatible_generalization/semantic_selection_control.md	supporting_paper_source	Semantic Selection Control for Structure-Compatible Generalization
papers/structure_compatible_generalization/semantic_selection_control_preregistration.md	preregistration_or_runbook	Semantic Selection Control Preregistration
papers/structure_compatible_generalization/structure_compatible_generalization.md	supporting_paper_source	Structure-Compatible Generalization
papers/two_bottlenecks/paper.md	primary_paper	Two Bottlenecks: DeltaE Auxiliary Bootstraps XOR Valence Geometry When Decoupled from Sparse-Reward Policy Gradient
papers/valence_object_formation/paper.md	primary_paper	Objects Form from Concern: Valence-Coupled Encoders Cluster the World by Causal Role, Not Sensory Similarity
papers/valence_tapestry/paper.md	primary_paper	Tapestry of Valence: Vector DeltaV Models Adapt Medicine Handling Under Internal-Weight Shifts; Scalar Drive Models Cannot
papers/vector_first_order_self/paper.md	primary_paper	Vector First-Order Self: Autonomous Identifying Interventions for Multi-Valence Homeostatic Agents
papers/vector_first_order_self/preregistration.md	preregistration_or_runbook	Paper 20B - Pre-Registration
papers/viable_computational_bodies/paper.md	primary_paper	Viability-Guided Evolution of Syntax-Bearing Computational Bodies
papers/viable_computational_bodies/preregistration.md	preregistration_or_runbook	Paper 2B Pre-Registration
papers/virtual_governor_stress_signal/README.md	readme	Virtual-Governor Stress Signals Paper
papers/virtual_governor_stress_signal/paper.md	primary_paper	Virtual-Governor Stress Signals for Local Action Recovery
papers/weakness_invariance_neurips/paper.md	primary_paper	Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization
papers/weakness_invariance_neurips/roadmap.md	supporting_paper_source	Weakness Invariance Paper Roadmap
papers/world_responds/paper.md	primary_paper	When the World Responds: Action-Related Shocks and the Limits of Null-Anchored Self/World Attribution
papers/world_responds/preregistration.md	preregistration_or_runbook	Paper 22 - Pre-Registration